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Review of Temporal Reasoning in the Clinical Domain for Timeline Extraction: Where we are and where we need to be

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ABSTRACT

Understanding a patient's medical history, such as how long symptoms last or when a procedure was performed, is vital to diagnosing problems and providing good care. Frequently, important information regarding a patient's medical timeline is buried in their Electronic Health Record (EHR) in the form of unstructured clinical notes. This results in care providers spending time reading notes in a patient's record in order to become familiar with their condition prior to developing a diagnosis or treatment plan. Valuable time could be saved if this information was readily accessible for searching and visualization for fast comprehension by the medical team. Clinical Natural Language Processing (NLP) is an area of research that aims to build computational methods to automatically extract medically relevant information from unstructured clinical texts. A key component of Clinical NLP is Temporal Reasoning, as understanding a patient's medical history relies heavily on the ability to identify, assimilate, and reason over temporal information. In this work, we review the current state of Temporal Reasoning in the clinical domain with respect to Clinical Timeline Extraction. While much progress has been made, the current state-of-the-art still has a ways to go before practical application in the clinical setting will be possible. Areas such as handling relative and implicit temporal expressions, both in normalization and in identifying temporal relationships, improving co-reference resolution, and building inter-operable timeline extraction tools that can integrate multiple types of data are in need of new and innovative solutions to improve performance on clinical data.

1. Introduction

Temporal reasoning is the ability to extract and assimilate temporal information to reconstruct a series of events such that they can be reasoned over to answer questions involving time. Medicine is one such area where temporal reasoning is vitally important to the care of patients as it plays a significant role in diagnosis and treatment decisions. The frequency or duration of symptoms could mean the difference between being diagnosed with a given disease or not, or the need to undergo a procedure. For example, if a child tests positive for strep throat seven times in a single year, the physician may consider performing a tonsillectomy [1]. Additionally, medications are taken at various frequencies, and tests may or may not be performed within certain time frames of symptoms. All of this information is stored in a patient's Electronic Health Record (EHR), and every aspect of EHR data contains some type of temporal component.

Prior to diagnosing or treating a patient, physicians and other medical care providers must understand the patient's medical history in order to know how to best care for them. This involves reconstructing the timeline of medically relevant events leading up to the patient's present condition, including the progression of symptoms, when tests or medical procedures were or are to be performed, and when medications were taken and for how long. All of this data is present in the patient's EHR in both structured (diagnostic codes, medications lists, procedure dates, etc) and unstructured (clinical notes, pathology reports, discharge summaries, etc) forms. One task performed by physicians prior to seeing a patient is the review of that patient's medical history documented by the EHR. This task can be time consuming and incomplete; however, it has been shown that visualizing longitudinal clinical data reduces the time it takes for medical professionals to assimilate a patient's information and assess their health status [2,3]. Having the ability to automatically extract and visualize a patient's medical timeline would allow medical professionals the ability to grasp the patient's condition more quickly and completely without having to read through and digest potentially large numbers of medical notes and EHR data prior to providing care.

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There have been many systems built to visualize a patient's EHR using structured data such as clinical encounter dates, admission/discharge dates, or dates of diagnoses or procedures (for example, [4–6]). However, temporal information contained within unstructured notes and reports is often ignored due to the inaccessibility of the information, even though these notes are the most frequently viewed information by physicians when reconstructing a patient's history [7]. Clinical notes contain medically relevant information about the patient's condition, including patient histories, which should be integrated into visualizations of a patient's medical timeline. Unfortunately, there are a small number of systems that can currently do this with unstructured clinical texts (see Section 7).

The field of Clinical Natural Language Processing (NLP) focuses on the extraction of information from unstructured clinical texts. Clinical texts are difficult to parse due to a variety of idiosynchronicities such as medical jargon, shortened, non-grammatical sentences, department and even physician specific formatting, and medical abbreviations that can mean something different in different medical contexts. These texts are also rich in temporal information as everything in medicine revolves around time. Clinical Timeline Extraction is an area of Clinical NLP that specializes in connecting temporal information in clinical texts to clinical events to form a timeline that can be reasoned over and visualized. This is a high-level task that relies on the good performance of several lower-level NLP tasks, namely 1) temporal expression recognition and normalization, 2) clinical event identification, 3) temporal relation identification, 4) clinical event co-reference resolution, 5) event ordering, and 6) timeline visualization. There has been a massive amount of work done on each of these tasks over the past several decades with the ultimate goal of performing temporal reasoning, which requires an events timeline. However, performance of current state-of-the-art timeline extraction pipelines are still not good enough to integrate into clinical practice leaving many areas of progress open to new and innovative ideas.

This work reviews recent progress in the area of timeline extraction from unstructured texts in the clinical domain and highlights specific areas in need of further work for each of the timeline extraction tasks listed above. In this review we first provide a high-level overview of the history of temporal reasoning in Clinical NLP in Section 2, followed by a description of the types of temporal information found in clinical texts in Section 3, and how this information is annotated for computational use in Section 4. Section 5 is an overview of the temporally annotated clinical corpora and shared tasks that have driven recent progress in this area. In Section 6 we provide details of each step involved in the generation of a clinical timeline from unstructured text along with a review of the current progress for each. Finally, Section 7 reviews the few end-to-end clinical timeline extraction pipelines for unstructured clinical texts and Section 8 discusses areas in need of improvement for timeline extraction systems.

2. A brief history of temporal reasoning

The history of temporal reasoning with NLP started in the 1950's and was focused on general domain texts. Much of the work through the early 2000's focused on the linguistics behind the temporal nature of discourse, how to represent temporal information (such as with intervals or discrete points in time), temporal named entity recognition, the development of annotation standards for temporal information, and the temporal ordering of events [8]. While there is much to discuss with respect to general domain temporal reasoning, this work is focused on clinical temporal reasoning, so we refer the reader to other reviews for a history of temporal reasoning in the general domain [9–12].

Clinical temporal reasoning has been around since the 1980's, but had trouble gaining traction with main-stream temporal reasoning NLP researchers due to the lack of access to a gold-standard clinical corpus. In 2012, Informatics for Integrating Biology and the Bedside (i2b2) was the first to release a fully de-identified and temporally annotated clinical corpus to the community as part of their 2012 challenge on evaluating temporal relations in clinical texts [13], followed by the release of the THYME (Temporal Histories of Your Medical Events) corpus for the Clinical TempEval challenge of 2015 [14,15]. For the first time, the NLP community at large had access to temporally annotated clinical corpora as gold standards for determining algorithm performance. This fueled the progress in temporal reasoning on unstructured clinical texts in the areas of temporal expression recognition and normalization, clinical event identification, and temporal relation classification. From the shared tasks that utilized these corpora, rule-based systems that expanded the top general-domain temporal taggers, like HeidelTime [16] and SUTime [17], emerged as performing the best for temporal expression recognition and normalization while statistical machine learning methods outperformed rule-based methods for event recognition and temporal relation classification tasks [18]. However, despite the boost in progress, temporal reasoning over clinical texts still posed several challenges, including the normalization of vague, relative, or implied temporal phrases; clinical event co-reference resolution; deciphering acronym and anaphoric expressions; identifying candidate temporal relationships; relative time anchoring; the end-to-end construction of a medical timeline from multiple documents; and the incorporation of structured EHR data with clinical narrative information [8,18]. In this review we focus on progress made after the landmark 2012 i2b2 Challenge in the area of clinical timeline extraction and refer the reader to several previous reviews of clinical temporal reasoning in the clinical domain with progress made prior to this challenge [8,18–20].

3. Temporal information

Temporal information in text conveys information about the passage of time or specific points in time. The basic units of temporal information are dates and/or times (e.g. April 4, 2020, 11:45 pm), and all expressions of temporal information in text are ultimately distilled down to some combination of dates and times that describe when, or for how long and event occurred. However, temporal information is not always expressed in such an explicit fashion. According to Lim et.al. [12] there are five ways temporal information can be conveyed in text: explicit, implicit, relative, vague, and non-consuming.

Explicit temporal information directly relays the basic temporal units of dates and/or times, such as "February, 18, 2020" or "9am".

Implicit temporal information can be either globally implicit (relying on some global knowledge base of historical events such as "New Years Eve") or locally implicit (relying on information given elsewhere in the current document). With clinical data, globally implicit references could also be those referring to other medical events in a patient's record that are not included in the document being processed. For example, a patient could have had knee surgery in the past, but the current clinical note being processed only indicates "the patient fell 2 months after knee surgery". In order to figure out when the patient fell, we would need to know when the surgery took place, which is part of the global knowledge about this patient's medical history. On the other hand, locally implicit references are those that refer to explicit temporal information elsewhere in the same document. For example, the phrase "the patient fell on March 3rd" may precede the statement "the patient reported severe back pain 2 days after she fell". The phrase "2 days after" is in reference to when the patient fell, which was explicitly stated in the text previously.

Relative temporal information is either anchored to some event or the document time. Phrases such as "3 days ago" may be relative to the document creation time, whereas phrases such as "2 h after taking medication" are relative to when the event "taking medication" occurred whether or not the absolute time the patient took medication is known.

Vague temporal information convey estimates of when something happened, but is unable to be converted into specific dates and times.

The phrase "in the early 1980's" represents an unknown period of time presumably in the first few months of the year 1980; however, the start date, end date, and duration are vague and unknown. Similarly, clinical notes can contain phrases such as "the pain started about 2 months ago". The phrase "about 2 months ago" is both relative to the document time as well as vague due to the word "about".

Finally, Lim, et.al. [12] described *non-consuming temporal information*, which is temporal information that is not explicitly stated in the text, but is assumed to be provided or is general knowledge. This generally represents the document creation time, of which many of the relative references may be anchored to.

Being able to define the types of temporal information is great, but if the computer can't process or utilize that data it is useless for NLP. Thus, it is necessary to define *temporal annotation schemes*, which format temporal information from text in a way that is more easily processed by computers for temporal reasoning tasks.

4. Temporal annotation

Annotation schemes are used to normalize unstructured information in texts to a computer-readable format for downstream processing. Temporal annotation schemes are specifically designed to normalize temporal information and related events into a standard format that can be utilized for temporal reasoning tasks. This includes providing standardized formats for temporal expressions, events, and the many different types of relationships between them. How an annotation scheme is defined has a major influence on how temporal and event information in text is processed and interpreted. Temporal annotation schemes have evolved over the past 30 years from having a simple temporal value attribute for annotated events (Message Understanding Conferences of the 1990's), to specifically annotating temporal expressions with an expanding set of attributes and adding tags for events and temporal links (TIDES and TimeML), to specialization for the clinical domain (THYME-TimeML), and the development of an interval and semantic-based temporal schema (SCATE). Fig. 1 provides a high-level summary of the evolution of temporal annotation schema with details being discussed in the following sub sections.

4.1. Translingual Information Detection, Extraction, and Summarization (TIDES)

TIDES was first developed in 2000 after the Defense Advanced Research Projects Agency (DARPA) sponsored Message Understanding Conference (MUC) challenges to provide more semantic details to the temporal expression (TIMEX) tags used in previous challenges. To differentiate, TIDES refers to temporal expression tags as TIMEX2 tags. This annotation focuses on temporal expressions only, and are explicit in what can and can't be annotated with TIMEX2 tags. Any temporal expression that has enough information to be pinned to a timeline is considered markable in the TIDES schema. Tag attributes capture semantic information about the expression as well as its ISO-8601 normalized value. Table 1 shows a few TIMEX2 tags for different temporal expressions (examples taken from [21]). Tag attributes include the normalized value of the expression (VAL), if a temporal modifier exists (MOD), the anchoring date and time (ANCHOR_VAL), the relative direction of the anchor value (e.g. before or after) with respect to VAL, and whether or not this TIMEX2 is annotating a set of temporal values, such as a frequency (SET). In general, vague expressions that do not have enough information to identify a spot on a timeline are not markable by TIDES [21]. This includes sequencing and ordering expressions like "subsequent", manner adverbs such as "immediately", non-quantifiable durations like "permanently", negatives and non-existent times (e.g. "no time"), the token "time" when it refers to a situation or occasion such as "at this time", and frequency expression without a quantifier (e.g. "frequently" or "too often"). As the TIDES schema is focused only on temporal expressions, the schema does not include tags for events or relationships between events and temporal expressions. These tags are added in the next temporal annotation schema, TimeML.

4.2. TimeML

In 2003, a new schema named Time Markup Language (TimeML) [22] was released that defined a new TIMEX tag named TIMEX3. The TIMEX3 tag is based off of the the original TIMEX tag [23] and the TIMEX2 standard from TIDES [21]; however, this new schema addresses the issue of not annotating events and relationships between temporal expressions and events, which are key components to understanding the temporal nature of a text.

While the TIMEX3 tag is based off of previous TIMEX tags, it's composition makes it difficult to convert TIMEX2 annotations to TIMEX3 due to the added and altered attributes [24]. In the TimeML schema, the TIMEX3 tag is used for explicit time expressions including dates, times, and durations, and has more attributes than TIMEX2 that provide more semantic information about each expression. These include the type (DATE, TIME, DURATION, or SET), beginPoint, endPoint, a quantifier such as "every", a frequency such as "2X", the document function (e.g. CREATION_TIME, PUBLICATION_TIME, EXPIRATION_TIME, etc), if this is a temporal function (true or false), a value, modifiers, and an anchor time.

The TimeML schema also includes annotation tags for additional

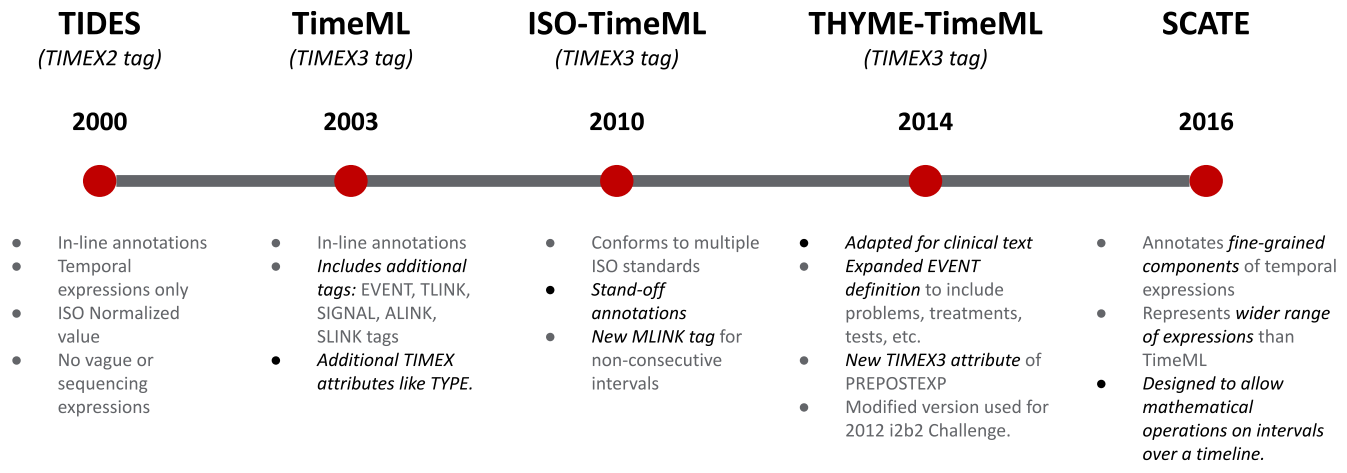


Fig. 1. Timeline of temporal annotation schema.

Table 1
Example temporal expressions with TIMEX2 annotation.

I returned to work at *twelve o'clock January 3, 1984*.
I returned to work at <TIMEX2 VAL="1984–01-03T12:00"> twelve o'clock January 3, 1984</TIMEX2>.
There has been a lot of rain *the past three weeks*.
There has been a lot of rain < TIMEX2 VAL="P3W" > the past three weeks</TIMEX2>.
She has been at work for *more than a month*.
She has been at work for <TIMEX2 VAL="P1M" MOD="MORE_THAN">more than a month</TIMEX2>.
Two years ago, the dance club drew about 100 students *each week*.
<TIMEX2 VAL="1997">Two years ago</TIMEX2>, the dance club drew about 100 students <TIMEX2 SET="YES" VAL="1997-WXX">each week</TIMEX2>.

entities that are not temporal expressions. These include the EVENT, LINK, and SIGNAL tags. EVENT tags annotate events (things that happen or occur) that need to be placed on a timeline. The SIGNAL tag is used to annotate function words that indicate how two temporal objects (TIMEX or EVENT) are related to each other (e.g. "when", "in", and "after") where previously these tokens were not mark-able by TIMEX2. LINK tags are of one of three types: TLINK, SLINK, and ALINK. TLINK tags are used to annotate a temporal link/relationship between an event and temporal expression (EVENT-TIMEX), two events (EVENT-EVENT), or two temporal expressions (TIMEX3-TIMEX3). Each TLINK can be one of 13 different types defined by the "relType" attribute, such as before, after, and identity. Finally, the SLINK (subordination link) and ALINK (aspectual link) tags are also relationships tags, and are used to annotate specific types of relationships between two events, or between an event and a signal (EVENT-SIGNAL). SLINKs must be one of 6 different types, and ALINKs are one of 4 types; thus, there are a total of 23 different types of temporal relationships that could possibly be annotated making LINK annotation in the TimeML schema a very complex and challenging task. Example TimeML annotations can be found in Table 2.

Table 2
Example temporal expressions with TimeML annotations.

I **returned** to work at *twelve o'clock January 3, 1984*.
I <EVENT eid="e1" class="OCCURRENCE">returned</EVENT> to work at <TIMEX3 tid="t1" type="TIME" value="1984–01-03T12:00">twelve o'clock January 3, 1984</TIMEX3>.
There has been a lot of **rain** *the past three weeks*.
There has been a lot of <EVENT eid="e1" class="OCCURRENCE">rain</EVENT> <TIMEX3 tid="t2" type="DURATION" value="P3W">the past three weeks</TIMEX3>.
She has been at **work** for *more than a month*.
She has been at <EVENT eid="e1" class="OCCURRENCE">work</EVENT> for <TIMEX3 tid="t3" type="DURATION" value="P1M" mod="MORE_THAN">more than a month</TIMEX3>.
Two years ago, the dance club **drew** about 100 students *each week*.
<TIMEX3 tid="t4" type="DURATION" value="P3Y">Two years ago</TIMEX3>, the dance club <EVENT eid="e1" class="OCCURRENCE">drew</EVENT> about 100 students <TIMEX3 tid="t5" type="SET" value="P1W" quant="EACH" freq="1w">each week</TIMEX2>.

4.3. ISO-TimeML

In 2010, the TimeML scheme was modified to provide an interoperatable temporal annotation scheme that conforms to the ISO standards ISO 24610–1:2006 FSR, ISO DIS 24611 MAF, and ISO DIS 24612 LAF [25]. The primary change was the move from in-line annotations to stand-off annotations where the text being processes is not altered (Table 3). The ISO-TimeML standard also introduces a new MLINK tag, which has the type "MEASURE" and is associated with durations. Previously, durations were treated as consecutive intervals; however, this is not always the case. For example, in the phrase "Sam taught for 4 h", it is ambiguous if Sam taught a consecutive 4 h or for a total of 4 h with breaks in between. The new MLINK relation allows one to identify this ambiguity in the annotation. Additionally, the ISO-TimeML standard provides a mechanism for proper counts of recurring events, where the previous annotation for the phrase "Sam taught every Tuesday in December" was unclear if the event "taught" occurred once or multiple times, the new ISO-TimeML provides a distributive mechanism to properly count the number of teaching events as four.

4.4. THYME-TimeML

Due to the unique challenges of temporal information extraction in the clinical domain, Styler, et.al. [15] developed the THYME-TimeML guidelines in 2014 for annotating temporal information from clinical texts. In THYME-TimeML, the definition as what qualifies as an event is expanded to include anything that would occur in a patients clinical timeline and is clinically relevant, such as diagnoses, tumor, illness, or procedure [15]. While the concept of EVENT was expanded, the attributes required were both simplified and expanded to be more relevant to clinical documents. For example, a new attribute for "contextual modality" was added that includes the values ACTUAL, HYPOTHETICAL, HEDGED, and GENERIC. Additionally, a new type of temporal expression is added to the TIMEX3 tag of PREPOSTEXP, which refers to clinically relevant and temporally complex terms such as preoperative, postoperative, and intraoperative.

Due to the higher frequency of temporal information in a clinical note or document, the number of temporal relations that needed to be annotated was large. Styler, et.al. were concerned about consistency in annotation and aimed to reduce the number of necessary TLINK annotations. They created the concept of a narrative container that was relative to the document creation time. Instead of creating TLINKS for all possible events, all events were assigned one of 5 container-based temporal relationship types: CONTAINS, BEFORE, OVERLAP, BEGINS-ON, and ENDS-ON. This effectively reduced the number of temporal relations needing to be annotated from 23 in the original TimeML specifications down to only five [15]. Both EVENTs and TIMEXs can be used as anchors for a narrative container. The advantage of this approach is that events are placed within an explicit temporal bound and it is not necessary to create all possible TLINKs to identify whether an event comes before, after, or during another event. Thus, with this change the CONTAINS relation is now the most frequent out of the previous relation types of BEFORE, OVERLAP, BEGINS-ON, and ENDS-ON. Due to the clinical relevance of the THYME-TimeML scheme, a simplified version was used as the basis of the 2012 i2b2 temporal challenge [13] annotation framework.

4.5. SCATE

The SCATE Schema (Semantically Compositional Annotations for Temporal Expressions) was developed by Bethard, et.al. [26] to address some of the shortcomings of the ISO-TimeML standard [25]. Specifically, ISO-TimeML has trouble representing time intervals that do not map to a specific calendar date, such as "2 summers ago", temporal expressions can only be relative to other times, and not events as in "four days postoperative", and the flattened structure of ISO-TimeML

Table 3
Example temporal expressions with TimeML-ISO stand-off annotations.

I **returned** to work at *twelve o'clock January 3, 1984*.
<EVENT eid="e1" start = 3 end = 11 class="OCCURRENCE"> <TIMEX3 tid="t1" start = 23 end = 53 type="TIME" value="1984–01-03T12:00">
There has been a lot of **rain** *the past three weeks*.
<EVENT eid="e1" start = 25 end = 29 class="OCCURRENCE"> <TIMEX3 tid="t2" start = 30 end = 50 type="DURATION" value="P3W">
She has been at **work** for *more than a month*.
<EVENT eid="e1" start = 17 end = 21 class="OCCURRENCE"> <TIMEX3 tid="t3" start = 25 end = 42 type="DURATION" value="P1M" mod="MORE_THAN">
Two years ago, the dance club **drew** about 100 students *each week*.
<EVENT eid="e1" start = 32 end = 36 class="OCCURRENCE"> <TIMEX3 tid="t4" start = 1 end = 15 type="DURATION" value="P3Y"> <TIMEX3 tid="t5" start = 56 end = 65 type="SET" value="P1W" quant="EACH" freq="1w">

annotations removes the compositional structure of temporal expressions. Thus, SCATE was developed to annotate the fine-grained components of temporal expressions, to represent a wider variety of temporal expressions, allowing for events to act as anchors, and using mathematical operations over a timeline to define the semantics of each annotation. Fig. 2 demonstrates the differences between SCATE annotation and that of the ISO-TimeML annotations.

5. Clinical temporal reasoning shared tasks and corpora

Shared Tasks provide a centralized and structured platform for advancing specific areas of research in NLP. An overview of the shared tasks in the general and clinical domains that include some type of temporal component is shown in Fig. 3. Temporal Reasoning and Information Extraction first appeared in NLP shared tasks in the 1990's with the Message Understanding Conferences (MUC 5, 6 and 7) [27–29]. These early tasks, however, were not focused on temporal information extraction, but rather identifying the temporal value of an entity, if present, as part of a named entity identification task. Temporal information extraction did not become the focus of a shared task until 2004 in the ACE TERN challenge [30], followed by a series of TempEval challenges [31–33]. The TempEval challenges focused on identifying and normalizing temporal expressions and identifying a variety of temporal relations, and were foundational in developing temporal reasoning systems in the general domain. Lim, et.al. [12] provides a good overview of these challenges and their contributions to the field. In this section, we focus on the temporal challenges and corpora in the clinical domain.

5.1. 2012 i2b2 challenge and corpus

Clinical temporal information extraction and reasoning shared tasks did not appear until 2012 with the Informatics for Integrating Biology and the Bedside (i2b2) Clinical Temporal Relations Challenge [13]. This task provided the NLP community with 310 de-identified discharge summaries from the MIMIC II (Medical Information Mart for Intensive Care) database out of the Beth Israel Deaconess Medical Center [34] with gold standard temporal annotations, and included 3 tracks: 1) temporal expression (TIMEX) and clinically relevant event identification (EVENT), 2) temporal relation identification (TLINK), and 3) end-to-end system. Unlike previous general domain temporal shared tasks, the i2b2 challenge narrowed events to those that were clinically relevant, namely clinical concepts (i.e. problems, treatments, and tests), clinical departments, evidentials, and occurrences. Temporal expressions were annotated with a modified version of the THYME-TimeML schema and included dates, times, durations, and frequencies with absolute values normalized to the ISO-8601 standard. Temporal relations consisted of all possible relations between two TIMEXs, two EVENTs, or a TIMEX and EVENT. Relation types consisted of BEFORE, AFTER, SIMULTANEOUS, OVERLAP, BEGUN_BY, ENDED_BY, DURING, and BEFORE_OVERLAP. Ultimately, 18 teams officially participated in the challenge; however, in the years after several other systems were published utilizing the same i2b2 corpus and further advancing the field [35–43].

5.2. TempEval challenges and THYME corpus

From 2015 to 2017, Bethard, et al. [14,44,45] hosted a series of Clinical TempEval challenges as part of SemEval following the task structure of previous TempEval shared tasks, only moving to the clinical domain. These challenges used the Temporal Histories of Your Medical Events (THYME) corpus [15], which is composed of 1,254 de-identified clinical notes and pathology reports from brain and colon cancer patients seen at the Mayo Clinic. The 2015, 2016, and 2017 challenges consisted of the same 9 tasks grouped into 3 categories: 1) TIMEX identification, 2) EVENT identification, and 3) TLINK identification. TIMEX types included DATE, TIME, DURATION, QUANTIFIER, PREPOSTEXP, and SET. Identified EVENTs were classified as ASPECTUAL, EVIDENTIAL, or N/A, and required properties to be annotated such as polarity and contextual modality. TLINKs were broken down into 2 main categories: DOCTIMEREL or CONTAINS. DOCTIMEREL relations were those where an EVENT has a BEFORE, OVERLAP, BEFORE-OVERLAP, or AFTER relation with the document creation time. CONTAINS relations were narrative container relations between EVENTs and TIMEXs. Participating systems were evaluated either as end-to-end systems, or were given TIMEX and EVENT annotations and judged on TLINK identification. Due to issues in getting participants access to the data in 2015,

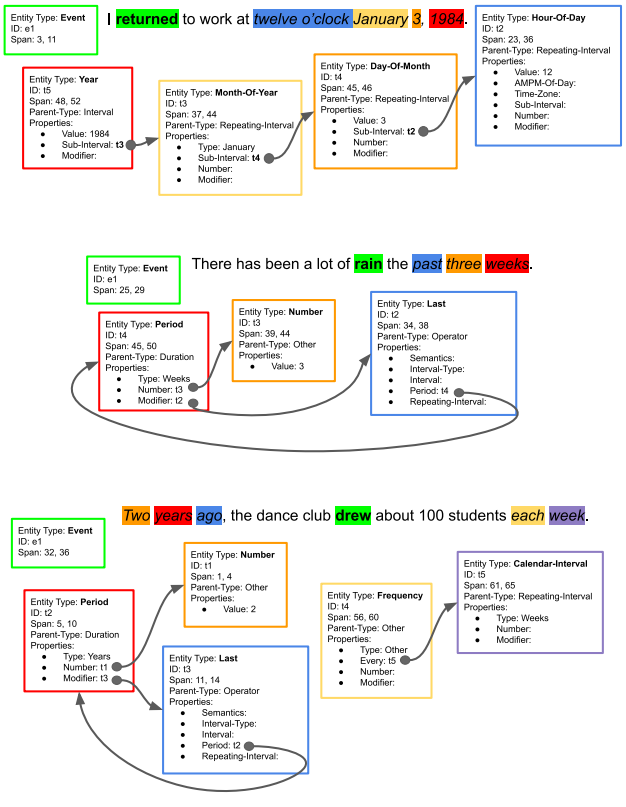


Fig. 2. Example SCATE annotations.

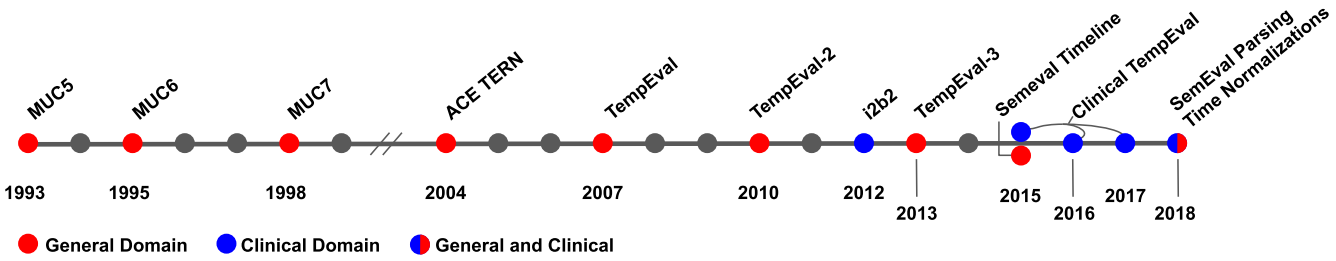


Fig. 3. Timeline of Shared Tasks that include Temporal Components.

the same challenge was run again in 2016 with 4 times as many participating teams. In both instances, teams using supervised machine learning approaches excelled in the TIMEX and EVENT identification tasks using a variety of features, indicating that these tasks are close to solved. However, the TLINK tasks proved to be challenging to all systems, especially the narrative container relations. In 2017, the same tasks were run again, however the aim was to address how well systems dealt with domain adaptation. Participating systems were given the colon cancer clinical notes and pathology reports for training, but were tested on the brain cancer cohort. All systems were reported to have a significant performance drop when tested on a new domain, which indicates that there is much work to be done in creating a generalizable timeline extraction system for clinical data.

The most recent temporal reasoning shared task was the SemEval 2018 Task 6: Parsing Time Normalizations [46] using the THYME corpus and the general domain AQUAINT News wire corpus. The goal of this task was to normalize fine-grained temporal information and relationships into the Semantically Compositional Annotations for Temporal Expressions (SCATE) schema developed by Bethard, et.al. [26]. This schema aims to improve upon the current TIMEX3/TimeML [22] standard by representing a wider variety of temporal expressions, allowing for events to act as anchors, and using mathematical operations over a timeline to define the semantics of each annotation. Two tracks were assessed: 1) Parsing text to time entities (event parsing and temporal relations were not assessed), and 2) production of time intervals. While over 40 teams registered for the challenge, only one team, Olex, et. al. [47], submitted results for comparison with the organizer's baseline system.

6. Timeline extraction

Timeline extraction is a high level temporal reasoning task that relies on the accurate performance of lower level Temporal Information Extraction (TIE) tasks. In order to reconstruct a useful and non-redundant medical timeline that can be used in clinical settings, the following steps must be implemented:

1. Temporal Expression Recognition and Normalization
2. Clinical Event Identification
3. Temporal Relation Identification
4. Clinical Event Co-Reference Resolution
5. Temporal Event Ordering
6. Timeline Visualization

In the following sections we define and review the current state of each of these step in the Clinical NLP realm along with a review of existing end-to-end clinical timeline extraction systems and the state of evaluating these systems.

6.1. Temporal expression recognition and normalization

Temporal expression recognition is the task of identifying which span of text contains a temporal expression (TimeML tag TIMEX3, referred to as a TIMEX). Temporal expressions can be annotated as one of four types in the TimeML schema: date, time, duration, and frequency. Each expression type can be composed of different types of temporal information such as explicit ("February, 18, 2020"), relative ("after the surgery"), implicit ("on Labor Day"), vague ("about 6 months ago"), or a combination ("about 2 weeks after Labor Day"). Non-consuming temporal information is generally not annotated as a temporal expression and is instead used to anchor relative expression to implicit information such as the document creation time. Each TIMEX is composed of one or more *temporal entities*, such as day-of-week, hour-of-day, month-of-year, etc. For example, the TIMEX "February 18, 2020" contains 3 temporal entities: month-of-year, day-of-month, and year. A TIMEX can also contain *modifiers* that indicate whether the temporal phrase is referring to the past, present or future (e.g. "last month" vs "this month" vs "next month").

Temporal expression normalization is the task of interpreting the value of an identified TIMEX and storing it in a computer-readable format. This format is generally the ISO-8601 standard, which is supported by the current temporal annotation schemes discussed in Section 4. In general, explicit dates and times, such as "May 4, 2020", are straightforward to normalize into the ISO standard. However, implicit, relative and vague TIMEXs are difficult to process due to the need for additional implicit information, which may be easily identifiable to a human reader, but not apparent to a computer.

Initial progress in TIMEX recognition and normalization came from the general domain with the TempEval-2 and 3 challenges in 2010 and 2013 [32], where TIMEXs were to be identified and normalized, including the identification of the type and value attribute of the TIMEX3 tag in the TimeML annotation scheme. In both of these challenges, the rule-based HeidelTime [16] tool achieved the best performance in identifying TIMEXs and their type. Other high performing systems included the data-driven ClearTK [48] system, which used SVMs, and the rule-based SUTime [17]. However, while all of these tools performed well on identifying a TIMEX span and type, it is clear from all the teams in both challenges that assigning the correct value of the TIMEX was a difficult problem indicated by the drop in performance.

TIMEX recognition and normalization moved to the clinical domain with the i2b2 2012 Temporal Challenge [13]. The top performing system for the TIMEX task was MayoTime, a rule-based system from the Mayo Clinic [49] that was built off of the top TempEval performer HeidelTime. All of the other systems were either rule-based or hybrid methods, with several others also integrating HeidelTime into their workflow. Interestingly, the HeidelTime team also participated in the i2b2 challenge; however, the tool trained for general-domain temporal information extraction performed poorly on the clinical data. This is likely due to the added challenges of parsing text that is temporally dense and contains many more frequency-based expressions than general domain text [50]. However, teams that integrated HeidelTime in with additional rules to compensate for the added clinical challenges performed better [13]. While the top performing systems did well on identifying the span, type, and modifier for the TIMEX3 tag, they all still

saw a drop in performance when it came to identifying the normalized value. Since the i2b2 challenge, rule-based systems are by far the preferred method for the task of identifying and normalizing TIMEXs from clinical text [38,43,51–53] followed by hybrid [54,55] and purely data-driven [56,57] approaches.

6.2. Clinical event identification

Event identification (TimeML tag EVENT) is the task of identifying things that happen and are of importance in a text. The definition of what qualifies as an EVENT can differ depending on the task at hand. In the general domain, an event is usually defined as a situation, action, or state of being [10], and is generally represented by syntactically inflected verbs or event nominals. In the medical domain, events are restricted to medically-related events such as tests, treatments, and medical problems [13,58]. For example, in the phrase "This morning the patient felt pain in her back when standing." the verbs "felt" and "standing" would be annotated as events in the general domain, whereas "pain" is the event of interest in the medical domain. Additionally, events always have a temporal component, such as when they occurred or for how long. In the above example, the temporal component of the medical event "pain" is "This morning". Identifying the temporal components of events is especially important in the medical domain as the frequency, duration, or order of symptoms may play a role in developing a diagnosis or treatment plan; thus, it is vitally important to identify these temporal aspects of medical events so that they can be correctly ordered on a timeline.

The 2010 and 2012 i2b2 challenges defined a clinical event as a clinical concept (i.e. problem, treatment, or test), clinical departments, evidentials (i.e. identifying the source of information), and occurrences (i.e. events like "admission" or "transfer" that happen to a patient) [13,58]. While this definition may cover most events of interest, it may not cover all relevant events depending on the task at hand. For this reason, Dehghan [54] expanded the definition of event to also include health and quality of life indicators, which are relevant when constructing a timeline of important events for a patient with a disease that causes long-term mental and emotional health issues such as childhood central nervous system (CNS) cancer. Thus, it is important to realize that the definition of event is task-specific, so building a catch-all classifier may be challenging.

Clinical event identification is similar to the classic Named Entity Recognition (NER) NLP task, except we are only looking for certain medically-related types of entities. In both the 2010 and 2012 i2b2 challenge tasks on clinical event identification, systems implementing conditional random fields (CRF) for event span detection, and support vector machines (SVMs) for event attribute classification performed the best for this task [58], however current research in both event and relation extraction have been exploring the use of deep learning-based approaches[59].

6.3. Temporal relation identification

Temporal relations (TimeML tag TLINK), are relationships between two entities that have a temporal component. TLINKs can exist between a TIMEX and an EVENT, between two EVENTs, between two TIMEXs, and between each EVENT and the document creation time (DCT). For example, in the phrase "a full hip replacement was performed on 6/22/92" the EVENT is "hip replacement" and the TIMEX is "6/22/92", where the TIMEX refers to when the event happened–this is an example of an *explicit relation*. EVENT-EVENT relations can also occur, for example, in the phrase "she became sick after visiting the store" there are 2 events, but no explicit temporal phrases. However, the event "became sick" is temporally linked to the event "visiting the store" with the term "before", which is an example of a *relative relation*. Finally, a third type of relation, the *implicit relation*, occurs when events are ordered. For example, in the passage "At 2 pm the patient was admitted to the hospital. They had taken their medication at 11am." there are 2 explicit relations stated, one for being admitted at 2 pm and one for taking medication at 11 am. There is also an unstated *implicit relation* between events where the patient took their medication *before* being admitted to the hospital.

Until the 2012 i2b2 challenge, not much attention had been paid to temporal relation extraction in the clinical domain [18]. The majority of the work was done in the general domain [9,11,12], and temporal relation extraction in the clinical domain posed new challenges. This included having to deal with the large increase of temporal expressions due to the increased temporal density of clinical notes versus general domain text [50], which creates more candidate relations needing to be filtered to the medically relevant ones. It also included dealing with implicit and relative relations where no explicit temporal expression is present in the text. Finally, clinical text is highly redundant [7,60] and requires sophisticated co-reference resolution techniques in order to build out non-redundant timelines of events.

The temporal relation track of the 2012 i2b2 challenge spurred a variety of machine learning and hybrid approaches to identifying TLINKs [13]. Those based on supervised machine learning approaches modeled the relation task as a pair-wise classification problem, where the algorithm had to classify relations between all pairs of events, temporal expressions, and the document creation time, regardless of whether or not they were actually related [36,42,49,61–64]. This strategy created n-squared TLINKs that had to be classified where most were a non-relation, which created a highly unbalanced set of training data. Some tried to reduce the number of relations by filtering to candidate relations [65], but this still left a lot of relations for classification. Hybrid approaches utilizing rule-based pair selection, CRFs, and SVMs performed the best; however, significant challenges still remained. While systems accurately identified relations between entities and the document creation time (DCT), all systems had difficulty narrowing the candidate entity pairs, anchoring relative temporal expressions, and identifying inter-sentence TLINKs, and these still remain areas in need of improvement.

Since the i2b2 challenge there has been some progress in TLINK identification for clinical texts. In the following subsections we review some solutions to handle inter-sentence TLINKs, strategies to target specific types of TLINKs, neural network approaches, and the contributions of BERT to temporal relation extraction.

6.3.1. Early inter-sentence temporal relation identification strategies

Inter-sentence TLINKs are difficult as they also require some level of coreference resolution (see Section 6.4). Cheng, et.al. [35] found their supervised machine learning classifier using MaxEnt performed poorly on these, so implemented a rule-base approach for inter-sentence TLINKs. D'Souza [37] found that many of the inter-sentence TLINKs were not actually annotated in the i2b2 corpus, so proceeded to augment the i2b2 annotations to compensate. Lin, et.al. [42] contributions were the first open source state-of-the-art end-to-end system that performed comparably to the top i2b2 system. It used the same features as previously published systems and implemented a multi-layered approach by identifying course-grained relations followed by intermediate and fine-grained.

6.3.2. Targeting specific temporal relation types

Many of the systems derived from the i2b2 challenge tried to identify both the explicit and implicit temporal relations simultaneously. This resulted in a lot of relations that needed to be identified. Lee, et.al. [39] argued that systems need to be good at identifying explicit temporal relations before identifying the implicit relations as they are based on the explicit temporal relations. Thus, Lee and colleagues [39–41] focused their work on defining, annotating, and identifying "direct" temporal relations, which are defined as explicit relations between an EVENT and TIMEX from within the same sentence. While their initial system based off of the top performing Vanderbilt system from the i2b2

challenge performed poorly [39], subsequent attempts that included the augmentation of machine learning features to include document format features, dependency features, and semantic roles along with previously used lexical, syntactic, and contextual features improved performance [40]. Viani, et.al. [59] also reduced the types of TLINKs identified in their work, but narrowed them to a specific medical condition instead of using syntactic properties like Lee, et.al. [39]. Viani, et.al. implemented a rule-based system to identify TLINKs that were relevant to the symptom onset timeline of patients diagnosed with schizophrenia. Because their system rule-set was built for a specific clinically oriented task for a specific medical condition, the developed rules and methodology are not generalizable, and many of the rules depend on their hospital system's specific format of clinical notes. Even with these highly specific constraints on the types of TLINKS targeted, and the manual annotation of events and TIMEXs for the corpus used, Viani, et.al.'s performance had an accuracy of 0.67, which is not much better than the more generalizable system produced by Lee, et.al. and Guan, et.al. [39–41].

6.3.3. Neural network approaches

Prior to 2017, temporal expression relation extraction was primarily implemented via a rule-base or hybrid approach with Support Vector Machines (SVMs) commonly chosen as the machine learning component using hand-crafted linguistic features [36,38,42,49,61–65]. However, in 2017 neural network architectures took the stage for the temporal relation extraction task. Dligach, et.al. [66] and Tourille, et.al. [67] were among the first to utilize neural networks for temporal relation extraction. Dligach, et.al. pitted the Long Short Term Memory (LSTM) units and Convolutional Neural Networks (CNN) architectures against each other using the THYME corpus from the 2015 Clinical Temp Eval Challenge [14] for identifying EVENT-TIMEX and EVENT-EVENT intra-sentence temporal relationships. Input was minimally processed raw tokens and/or POS tags with XML markup of the temporal expressions and events. CNNs were shown to outperform the LSTM architecture for the Clinical Temp Eval 2015 challenge. However, shortly thereafter, Tourille, et.al. [67] implemented a Bidirectional-LSTM (Bi-LSTM) that included character-level features for both intra- and inter-sentence temporal relations on the THYME data set with comparable performance on intra-sentence relations and superior performance with both inter- and intra-sentence temporal relations. Lin, et.al. 2018 [68] took the BiLSTM from Tourille, et.al. further by including a self-training RNN framework that utilized out-of-domain word embeddings to create a silver standard for training. This self-learning was also applied to the top performing, SVM-based THYME system. Lin, et.al. discovered that SVM-based systems are unable to learn from self training using a silver standard and identified that the SVM was just calling more instances as positive. The self training did improve the RNN-biLSTM and RNN-GRU models, and including the out-of-domain features improved cross-domain performance as well. However, Lin, et.al. is still only using intra-sentence relations, and results were not very different from Tourille, et.al.

6.3.4. The Era of BERT

In 2018, researchers from Google AI introduced Bidirectional Encoding Representations from Transformers (BERT) [69] as a new pre-trained language representation model. Representational Learning in NLP aims to learn informative numerical representations of words. These learned representations can take the place of, or augment, manually defined features that are fed into machine learning models for prediction tasks [70]. However, learning good and informative representations is a computationally expensive task and requires massive amounts of data, thus, there has been much attention paid to pre-trained models in recent years. Pre-training allows users to take advantage of the information in large corpora in the form of a pre-trained deep neural network with which they only have to fine-tune the top layer to obtain word embeddings specific for their task. BERT is one of these models and has taken the NLP community by storm, producing new state-of-the-art performances on many NLP tasks [69].

BERT differs from the classic distributed word embedding models, such as Word2Vec [71] and GLoVE [72], by learning contextualized embeddings. A *contextualized embedding* is one in which the context of a token is incorporated into the embedding vector. Distributed models use the co-occurrence of words to form their embeddings and thus, generate a single embedding for each word no matter how it is being used in a given sentence. On the other hand, BERT generates a slightly different embedding depending on how a specific token is used in a sentence. For example, the word "bank" can have multiple meanings depending on how it is used, such as in "The river bank was perfect for fishing." or "I visited the bank to withdraw some money.". Other pre-trained language models, such as ELMo [73] and OpenAI GPT [74], also generate contextualized embeddings using a unidirectional model; however, BERT implements bidirectional representations using a "Masked Language Model" where it is able to use the context both before and after a token, and thus, incorporates more information into its embedding.

Since the debut of BERT, it has been used in many NLP applications, including in the area of temporal reasoning. In 2019, Lin, et.al. [75] converted their Recurrent Neural Network (RNN) self-training system [68] to utilize BERT and account for cross sentence relations by implementing a window approach instead of sentence by sentence. The BERT base was fine-tuned on MIMIC and PubMed, and self-training using silver standard annotations was implemented similar to Lin, et.al. [68] except with the window method instead of sentence-based. Improvements in performance, however, were mild, increasing the F1 score on the colon cancer test set to 0.684 from the previous best of 0.629. Lin, et. al. also assessed the performance on the development set of the inter-sentence relations to get an F1 at 0.33, but was not able to surpass the performance of Tourille, et.al. [67] with an F1 of 0.482, indicating much work is still needed in this area. Interestingly, they found that the window-based method does not work well with the previous BiLSTM method due to the unique characteristics of BiLSTMs. One of the disadvantages of the BERT model is its computational complexity due to having to encode the same sequence n x (n-1)/2 times. In 2020, Lin, et. al. [76] converted this method to use a one-pass encoding mechanism inspired by Wang, et.al. [77], which achieves the same performance but with a significant reduction in computational complexity, reducing the training time by several hours to days depending on the data set size. More recently, Guan, et.al. [41] has produced the latest state-of-the-art in direct TLINK identification using RoBERTa [78], which is a multi-layer transformer encoder modified from the original BERT [69]. By training on more data, longer sequences, removing the next sentence prediction objective, and altering the masking pattern, Guan, et.al. [41] utilized BERT to predict the temporal relation presence and type using the direct temporal relation annotated corpus from Lee, et.al. [39]. However, similar to previous work, only intra-sentence relations were targeted that meet specific lexical-syntactic requirements for a direct temporal relation (defined in [39]).

Overall, there has been progress in identifying temporal relations from refining focus to explicit relations, to implementing neural networks, and using information contained in deep neural networks such as BERT; however, there is still work to be done. Strategies to improve performance in the area of inter-sentence relations and relations involving implicit temporal expressions are needed to move the area of TLINK identification forward.

6.4. Clinical event co-reference resolution

Coming from a non-linguistics background, deciphering the precise meaning of the term *co-reference* is surprisingly challenging as its definition is intertwined with the concept of *anaphora* in the NLP and computational linguistics literature. Recent reviews on the topic state that anaphora and co-reference resolution are two distinct yet overlapping tasks in NLP [56,79]. In this work we adopt the definitions of Sukthanker, et.al. [79] and Tourille, et.al. [56] for Co-reference

Resolution as the task of identifying equivalency classes among identified entities or events within a single, or across multiple, documents.

With regards to timeline generation, co-reference resolution is vital to building non-redundant timelines, especially when multiple related documents are processed. Clinical documents offer unique challenges to the task of co-reference resolution as compared to general domain texts. For example, clinical documents are highly repetitive and redundant due to the copying and pasting of content from previous documents or entries by clinicians to ensure the most relevant information is easily found [60,80]. This creates a greater number of entities that must be placed in an equivalence class. Also, the narrative structure of clinical documents is limited and frequently not present due to domain knowledge of how notes are written and should be interpreted, and clinician shorthand. Both of which can change across different clinicians, departments, and medical facilities. This means the same event or entity could be referenced in multiple ways that can differ dramatically across documents such as a patient's history to a radiology report that both refer to a similar medical event. On the flip side, event mentions that have identical lexical and syntactic properties could be referring to completely different real-word events; thus, context and lexical-syntactic properties alone are insufficient to determine if the events are equivalent. Finally, co-references can not only span sentences, but also document sections and across different documents in the patient's medical record. This results in different contexts surrounding each mention of an entity or event that need to be normalized and resolved before the entity or event can be placed in an equivalence class and then properly placed on a timeline.

There was little progress in clinical co-reference resolution until the 2011 i2b2 Co-reference Challenge [81], which expanded the focus beyond noun phrases containing anaphores and antecedents to clinically relevant persons, entities, and events. Using discharge summaries, entities were defined as one of PERSON, PROBLEM, TREATMENT, or TEST. Entities could only be co-referential if they were the same entity type, which helped to resolve the negative bias problem for training data. For example, the entities "temporal artery biopsy" and "that testing" in Quotation 1[1] are co-referent of the same medical procedure, and thus belong to the same equivalency class, whereas the entity "she" clearly belongs to a different class and does not have to be evaluated with the medical procedure entities. This entity type restriction was used by many of the participating systems to build separate modules for the resolution of each entity type. Overall, 20 systems participated in the challenge, and consisted of rule-based, hybrid, and supervised learning approaches. Uzuner, et.al. [81] notes that all systems, regardless of strategy, that incorporated external domain knowledge, such as the Unified Medical Language System (UMLS) [82] and Wikipedia, showed improved performance. Xu, et.al. [83] built the top performing system which uses a binary SVM to identify all of the PERSON mentions from the PROBLEM, TREATMENT, and TEST mentions, then trains SVM classifiers using a large list of contextual and world knowledge features to create PERSON co-reference chains (Primarily relating to the patient), and a separate classifier to identify the other types of co-reference chains. In the i2b2 challenge, Xu, et.al. obtained an F1 score of 0.915, which is very good for this task. While the 2011 i2b2 Co-reference Challenge encouraged progress in the area of clinical co-reference resolution, it focused on discharge summaries where clinical text is written in a more narrative fashion, and only required co-reference to be determined within a single document. Further work is needed to address co-reference resolution across multiple documents and in documents that have less of a narrative structure.

She was scheduled to receive a temporal artery biopsy, but she never followed up on that testing. (1)

Until recently, co-reference resolution had been performed without consideration of the temporal information. A few systems in the 2011 i2b2 challenge utilized times and dates as features to help distinguish events with similar contextual, lexical, and/or syntactical properties [83–85]. However, recent progress in clinical event co-reference resolution has determined that temporal information is vital to resolving event co-references both within a document and across multiple documents since event mentions can only co-refer if they happened in the same time frame [54,56,86,87]. Tourille [56] even argues that in the task of extracting clinical timeline, event co-reference and temporal information extraction should be performed jointly as they are complementary and connected.

Since the 2011 i2b2 Co-Reference Challenge, only a few papers have tackled incorporating co-reference resolution into timeline extraction. In 2014 Raghavan, et.al. [86] addressed the task of identifying co-referring entities/events across multiple documents by modeling it as a multiple sequence alignment. Clinical events from each document were ordered into a sequence, then contextual and temporal features were used to align events across documents. To this day, Raghavan, et. al. is the only one to address cross-document co-reference resolution. Around the same time, Dehghan, et.al. [88] performs event co-reference resolution by using the lexical similarity of the context surrounding the event using a SoftTFIDF score. Any event pair score that is greater than 0.8 is assigned a new TLINK with the category Overlap, then a transitive closure of all TLINKS is performed. Both of these methods integrate the event co-reference resolution task with the identification of temporal relations, and they do not consider co-reference relationships in the PERSON category as defined by the 2011 i2b2 challenge. As the main purpose is to build a timeline of events, PERSON co-references are irrelevant in this context as we are only concerned with medical events that happened to the patient. Finally, the most recent work that integrates temporal information into clinical co-reference resolution is Tourille, et.al. [56] where an event's temporal relation to the document creation time is included as a feature for a neural entity-based mention-ranking method.

It is clear from the limited set of published works since 2011 that clinical event co-reference resolution is far from a solved problem. With the goal of clinical event timeline extraction, co-reference resolution must be able to identify equivalence classes of events across multiple documents written in different styles and at different times over a patient's medical history. As of yet, only one group has attempted cross-document co-reference resolution [86], leaving much to be explored in this area. It is also clear that accurate clinical co-reference resolution requires the integration of temporal information. The type of temporal information that is the most useful (i.e. specific dates/time, relative relations, or relationships to a document creation time) is up for debate, but current evidence indicates that good clinical co-reference resolution needs temporal information and complete timeline extraction needs clinical co-reference resolution.

[1] Example from [81]

6.5. Temporal event ordering

The task of event ordering sounds, well, easy. Given explicit dates it is a simple matter of placing events chronologically on a timeline to induce order. However, when dealing with non-explicit, fuzzy, and relative temporal expressions, inducing order is a bit more difficult. The addition of timelines from multiple documents or data sources makes ordering events even more challenging, and hinges on the ability to perform event co-reference resolution within and across documents. In this work, we define the task of *event ordering* as the task of identifying the order of events while resolving sequencing conflicts (or not creating them to begin with).

When a set of temporal relations are viewed as a directed graph, a *sequencing conflict* occurs when a cycle is created; thus, any valid set of temporal relations should form a directed acyclic graph. This problem is closely related to identifying temporal relations, and in fact is a direct result of modeling the temporal relation task as a pair-wise classification task. These algorithms do not consider ordering of events when assigning relationship types like BEFORE or AFTER. Thus, when you try to put these events in order on a timeline you could run into a sequencing conflict that needs resolution. For example, suppose our temporal relation algorithm identified the following relations between events A, B, and C: A before B, B before C, and C before A. Through inference of the first 2 relations, we can infer A happened before C; however, the direct relation "C before A" states the opposite. Now the question is, which is right? When did event A happen with respect to event C? Resolving these cycles is part of the Event Ordering task.

To solve the ordering cycle paradox, Bramsen, et.al. [89] modeled pair-wise relations as an acyclic directed graph (DAG). To ensure no ordering cycles were created, they implemented a greedy strategy by retaining only the relations with the highest confidence from their relation classifier. After each new relationship is added, the DAG is expanded using transitive closure. If a cycle is detected, only the highest confidence relations are kept to maintain the DAG. While this strategy may not find a global maximum, the authors claim it is close enough. However, no attempt is made to see if this resolution strategy generates clinically correct timelines.

Raghavan, et.al. [90] takes a step back and looks at classifying events into course-grained time-bins based on their relation to the admission time (way before admission, before admission, on admission, after admission, after discharge) to create a course-grain ordering of events. A Conditional Random Field (CRF) is trained on temporal expression and narrative structure features such as sections, contextual bigrams, dictionary features, and explicit dates within the same sentences to place events into these time bins, which can then be used to identify a more fine-grained ordering of events if needed. Sarkar, et.al. [51] take a similar approach to order events in medical case reports where explicit temporal expressions are sparse, so one has to rely on relative temporal expressions. They first identify temporal breaks (i.e. current, past, future) in the narrative using a CRF. Ordering of events is then rule-based and is performed for each of the temporal contexts by ordering the sentences instead of individual events. Sarkar, et.al. is the first to explicitly focus on relative temporal expressions for event ordering in the clinical domain. Finally, Jeblee, et.al. [91] tackles the event ordering problem implicitly in their solution to temporal relation identification by forming it as a list-wise ranking problem instead of a pair-wise classification problem. They argue this method alleviates the temporal inconsistencies from pair-wise methods, eliminates the need to process n-squared relationships, and is easier for humans to evaluate. A linear neural network is used with contextual and event attributes as input features to rank events in a list-wise fashion. Viewing the temporal relation problem as a ranked list has the potential to more easily capture relative event relations as well, but this aspect was not addressed in this work and is noted as future work by the authors.

Temporal event ordering in the clinical domain is tightly associated with temporal relation extraction, but has not been paid much attention in the Clinical NLP field. Temporal inconsistencies are introduced, computational complexity increased, and human evaluation is difficult when modeling the temporal relationship identification task as a pair-wise classification problem. Modeling the temporal relation task as a list-wise event ranking problem resolves the issue with temporal inconsistencies and is easier for humans to comprehend and evaluate; however, current progress only uses simple relations and cannot model more complex or fine-grained event relations. Thus, event ordering is still an area of needed research and is tightly tied to the strategies used for temporal relation identification and classification. Methods are needed that can reduce the computational complexity of relationship identification and provide ordered sequences of events that are detailed and easily assimilated by medical professionals.

6.6. Timeline visualization

Timelines are most commonly visualized on a straight line with single points indicating when certain events occurred. Medical timelines, however, are more complicated due to the multiple layers of information that need to be represented, such as patient symptoms, procedures, and medications. In addition, durations of symptoms and medications are important to accurately visualize and cannot be represented as a single point on a line. While we may develop a highly sophisticated and accurate medical timeline extraction tool, it will be useless to clinicians in the field unless the information it has extracted can be graphically visualized in such a way that medical professionals can easily navigate events and quickly process and interpret the information to gain new insights about a patient's condition.

Medical timeline visualization is an active research field. In 2004, Martins, et.al. introduced KNAVE-II [92], a tool built on a distributed framework for visualization of and interaction with time-oriented clinical data. Pulling data from multiple clinical data sources and a medical knowledge base, KNAVE-II aids clinicians in answering time-oriented clinical questions about a patient. Timeline was developed by Bui, et. al. [4] and uses EHR structured data from multiple sources to provide a longitudinal, problem-centric view (such as cancer) of a patient's clinical data using multiple tracks for different data types. HARVEST [5] primarily utilizes the admission and discharge dates of patient visits to build a timeline that is displayed as a single linear track; however, it also analyzes clinical notes using NLP methods by normalizing disorders to UMLS concepts [82] followed by a topic analysis with a TF-IDF (Term Frequency-Inverse Document Frequency) matrix to identify the most informative disorders in a patient's record compared to all other patients. Glicksberg, et.al. [6] argued that the interoperability between developed timeline tools and ERH or data warehouse systems is a big problem for deploying a tool to multiple health care systems. They propose using a Common Data Model, such as OMOP[2], as the backdrop to a timeline tool, and developed a simplistic timeline visualization based off of structured EHR data. While many of the timeline visualizations only focus on a single patient, Kilmov, et.al. [93] expanded the KNAVE-II system to visualize and perform temporal reasoning over multiple patients for analysis and comparison. Likewise, Gotz, et.al. [94] developed a tool to aid clinical researchers in identifying temporal patterns present in multiple patients, where users define the concepts and outcomes they are looking for and then the patient's matching those with the timeline of relevant events is visualized for exploration and analysis.

All of the aforementioned tools are built to utilize EHR or data warehouse structured data and do not incorporate information in clinical notes that could be extracted by NLP. This means there may be a lot more clinically relevant information present in an EHR that is not easily accessible by medical professionals, even with current timeline visualization tools. Park, et.al. [95] argue that context and causality of events

[2] https://ohdsi.github.io/CommonDataModel/

are also important to represent on a timeline. They propose V-model, a temporal information visualization tool that can effectively visualize event causality, non-explicit temporal information, and multiple levels of temporal granularity for events. V-model exclusively uses clinical narrative texts to summarize patient histories on a timeline; however, it has only been evaluated on a small cohort of single documents.

While there are a variety of clinical timeline visualization tools available, there is still a need for an interoperable tool that can integrate multiple data types, including unstructured text, originating from multiple clinical systems into a comprehensible and intuitive visualization. This tool would need to be flexible, allowing for a global or problem-centric view of a patient record. The field is making progress on this front using structured EHR data; however, the incorporation of information for unstructured texts is lacking.

7. End-to-end timeline extraction systems

In this section we review end-to-end clinical timeline extraction systems currently published in the literature. We classify a system as an end-to-end timeline extraction system if it includes temporal expression recognition and normalization, event identification, temporal relationship identification, and either event ordering or event co-reference resolution for either a single document or over multiple documents. Any system that did not include methods or a tool to perform each step was not considered as end-to-end. This includes systems that used manually annotated temporal expressions or events [59,86], or those that used a pre-annotated gold-standard to obtain these entities [96,97]. It also includes systems that only utilize the structured temporal data in the EHR instead of extracting it from clinical notes [98–101], and all timeline extraction systems that only deal with general domain text [102–105]. Additional timeline extraction steps considered as a bonus are timeline visualization and performance evaluation (see Section 7.2 below).

7.1. End-to-end system review

Four published systems were found that met the end-to-end criteria for clinical timeline extraction (Table 4) [53,54,106,107]. Three of these systems use rule-based components [53,106,107], with the other using a hybrid approach by incorporating machine learning into the event and temporal expression recognition, and co-reference resolution steps [54]. Each system takes a slightly different approach to timeline extraction that is influenced by the type of timeline needing to be identified, such as differing definitions of what a medical event is, the type of temporal expressions targeted, the number of documents needing to be reasoned over, and the underlying framework.

Zhou, et.al. [106] and Jung, et.al. [107] were the first to develop end-to-end timeline extraction systems that were both extensions of previously built NLP systems. In 2005, Zhou, et.al. published the earliest complete timeline extraction system to integrate temporal information from clinical narrative reports into the existing MedLEE system. MedLEE is a rule-based NLP system that incorporates linguistic and clinical domain knowledge (e.g. UMLS, SNOMED, etc.) to annotate and structure clinical information into a frame-based representation. MedLEE's temporal tagger extracts temporal information into a Temporal Constraint Structure that defines the beginning and end of events. This data is then fed into the main NLP system to link events to temporal information. In post-processing the structured information and rules of discourse are used to model the timeline of events in a clinical narrative as a Simple Temporal Problem (STP). STPs are represented as a constraint graph, which effectively orders the clinical events based on their temporal constraints, and allows for non-explicit temporal relations to be inferred. At the time of publication, only a few basic rules had been implemented to perform co-reference resolution across multiple documents. MedLEE was designed to capture all medical events and thus, does not revolve around a specific use-case.

The next complete timeline extraction NLP system did not surface until 2011. Jung, et. al. [107] extended a general purpose NLP system [108] using deep natural language understanding (NLU), to extract medical concepts and related temporal information from cancer patient consultation notes. The core NLU system relied on previously developed, sophisticated parsers that employed several statistical and symbolic subcomponents, like POS tagging, and output information in Logical Form (LF), which is a frame-like semantic representation of the parsed text. Clinical domain and ontological knowledge was integrated into the LF representation from SNOMED and the UMLS. Clinical concepts, events, and temporal expressions were extracted using manually defined LF pattern-based rules. Relationship identification between events and concepts (where an event can contain many concepts) and between events and temporal expressions is rule-based and derived from the LF graph. Referential temporal expressions are resolved using LF information or the document time, and negations are considered to determine if a concept is present or not in the patient's medical history prior to visualization with the Simile Timeline Widget.

The systems built by both Zhou, et.al. and Jung, et.al were intended to be general purpose timeline extraction systems for clinical data, and both only focused on *single document narrative texts*. Thus, the methodology of each was geared towards parsing discourse structure, which may not be applicable to the often fragmented, highly specialized, and diverse types of clinical notes commonly found in the EHR. While Zhou, et.al. implemented some basic co-reference resolution across multiple documents, Jung, et.al. does not consider this step. Additionally, both systems only focus on using explicit temporal expressions, and neither system implements a metric for a formal performance evaluation.

In 2014, Deghan [54] developed a clinical timeline extraction system, referred to as "mining patient journeys", that addressed some of the short-comings of the systems from Zhou et.al. and Jung et.al. Namely, Deghan utilized multiple narrative documents (clinical and patient narratives), performed co-reference resolution across a patient record (i.e. across multiple documents), and evaluated the system's performance, both on each individual component as well as for the resulting timeline. However, the system developed by Deghan was built for the specific use case of identifying treatment timelines for survivors of childhood brain cancer.

Briefly, Dehghan used a hybrid approach with the EVENT, TIMEX, and TLINK components trained on the 2012 i2b2 standard. EVENTs included the i2b2 problems, treatments, and tests, in addition to health-related quality of life (HrQoL) concepts that were relevant to survivors of childhood brain cancer. A hybrid approach was taken to identify EVENTs and TIMEXs using Conditional Random Field models with lexical, syntactic, and orthographic features as input, along with rule-based approaches, such as regular expressions for formatted dates/times and dictionary lookup for HrQoL concepts. All EVENTs identified were mapped to UMLS concepts in order to utilize the knowledge base to easily identify high-level concept categories, which are used in co-reference resolution and timeline evaluation. An initial set of TLINKs were identified with a rule-base approach, then SputLink [109] was used to perform a transitive closure to identify all implied relationships, and to resolve any inconsistencies in temporal ordering. Only TLINKs that could be resolved to a point on a timeline were used in the final timeline construction phase (i.e. all relative or non-anchored TLINKs were removed). Co-reference resolution was performed within each document and across multiple documents in a patient's record. The intra-document co-reference resolution was based only on lexical similarity by using a SoftTFIDF score [110] on all candidate event pairs within a single document. Inter-document co-reference resolution was performed in a similar manner, but was limited to events assigned to high-level categories related to the specific use-case, such as surgery or radiotherapy. Finally, event ordering and visualization were done using applications referred to as "PathCluster" and "PathVisualization", which grouped events into high-level processes and assigned them to a time bin with a 6-month interval.

Table 4
End-to-End Clinical Timeline Extraction Systems

Tool:	MedLEE [106]	Deep NLU [107]	Dehghan [54]	Najafabadipour [53]
Year	2005	2011	2014	2020
TIMEX	Rule-based: Temporal Constraint Structure (TCS), explicit focus	Rule-based: Logic Form Parsing, explicit focus	Hybrid: CRF + Rules	Rule-based: Temporal Tagger
EVENT	Rule-based	Rule-based: LF Parsing	Hybrid: CRF + Rules	Rule-based: C-liKES and TNM Annotator
TLINK	Rule-based	Rule-based: LF Parsing	Rule-based	Hybrid: UDPipe + Rules
Event Normalization	Rule-based: absolute reference chosen over implicit	X	Rule-based	Rule-based: Semantic Similarity + Temporal Rules
Event Ordering	Temporal discourse, Simple Temporal Problems (STP) Graph	Assigns intervals to events	PathCluster	Rule-based: Selects earliest occurrence of unique events, sorts temporally
Visualization	X	Simile Timeline Widget	PathVisualization	X
Evaluation	X	X	Precision and Recall	P,R,F1 for components, manual for timeline system
Scope	Single Document	Single Document	Multiple Documents	Multiple Documents

Dehghan's approach to extract patient journeys is based on abstraction of low-level details to high-level processes and concepts. Ultimately, EVENTs are abstracted to higher-level concepts using the UMLS, and are then binned into 6-month intervals. The 6-month interval was chosen based on the use case as 6-months was the approximate follow-up time for the patients under study. Thus, within a 6-month time bucket, the exact order of events doesn't matter. Using a manually generated gold-standard, evaluation of the full timeline at the abstracted, high-level (i.e. "Oncology Treatment", "Oncology Diagnosis", etc.) resulted in 100% precision and recall; however, this high level information is not helpful to a treating physician. When details, such as type of chemotherapy, were required to match the gold-standard timeline, performance decreased. While Dehghan's system performs well for a specific use case, it still relies on narrative texts, and much of the co-reference resolution and performance evaluation are dependant on the high-level concepts defined by the specific use case. Additionally, the 6-month time-frame for binning events may not work well with diseases that are faster moving and have densely populated clinical notes. This system would most likely require detailed manual intervention in order for it to extract a timeline for a different type of disease or medical use-case.

The most recent clinical timeline extraction system is by Najafabadipour, et.al. in 2020 [53], and contains mostly rule-based components built over the UIMA framework [111]. This system was designed to extract the evolution of a lung cancer patient's health status over time starting at the date of diagnosis, and is the first timeline extraction system for Spanish clinical notes. Najafabadipour, et.al. define EVENTs as diagnosis concepts and tumor stages, and extract them using the C-liKES system [112] and TNM Annotator [113], respectively. TIMEXs are identified and normalized using the rule-based Temporal Tagger tool [114]. These annotations are then input into the Temporal Reasoning System, which identifies TLINKs using the UDPipe tool [115] to first build a dependency parse tree using a simple single layer neural network followed by a set of rules using syntax dependencies to determine if an EVENT is linked to a TIMEX based on the parse tree. The annotated EVENTs, TIMEXs, and TLINKs are stored in a structured database, which is used as input to the Timeline Constructor step. EVENT ordering and co-reference resolution are carried out during timeline construction. Capturing the evolution of changing health status requires a slightly different approach as you need to capture the *change in status* of a medical concept (e.g. tumor stage) that is associated with clinical events. This led Najafabadipour, et.al. to develop a unique inter-document co-reference resolution method that required 2–3 criteria be met for two events to be termed co-referential. First, events needed to be semantically similar. Unlike Deghan [54] who only used lexical cues, Najafabadipour, et.al. looks at the similarity between the assigned semantic classification of events. Second, the associated time for an event should be the same or contiguous after sorting. Finally, if an event is associated with a value (like tumor stage), a third criteria must be met where co-referential events must have the same value. This definition identifies continuing states of a patient, designed to identify the evolution of the patient's health status over time. Once events are ordered and identified as co-referential, the earliest event time is kept for that event and propagated to all co-referring instances. System generated timelines were compared to over 800 manually, expert-curated timelines. Errors were generated under one of two condition: 1) the system identified a different number of events, or 2) events had different date expressions from those identified by the experts. Overall, the system performed well under these metrics, but could be improved through annotation of negations and probabilistic terms. Najafabadipour, et.al.'s approach to timeline generation is specifically built for tracking the evolution of lung cancer patient status over time. This view on clinical timelines is novel and may make sense for many clinical conditions as a patient's status does change over time, and that information can be highly informative.

From the four end-to-end timeline extraction systems we can see that the methodology is influenced by the assumed type of text being processed and the use-case that is utilized. Because a patient's medical history is buried in multiple notes with multiple note types and grammar that is not always going to follow traditional rules, future timeline extraction systems should be flexible in processing this diverse data, as well as able to deal with the high level of redundancy in the EHR by integrating this data into a single contiguous timeline through robust co-reference resolution. In addition, timeline extraction systems need to focus on the ability to extract timelines relevant to any number of conditions or diseases that have any number of temporal pathways (e.g. a slowly progressing chronic non-terminal disease vs a quickly moving terminal disease vs a short-term illness such as a viral infection). The current tendency to focus on a single type of disease or cohort of patients, all with similar temporal progressions may lead to systems that do not generalize well for conditions with varying temporal properties. In addition, the extraction of temporal entities from clinical texts will generally be static across an EHR; however, the definition of a medical event needs to be standardized and applicable to all types of use cases, and robust methods to resolve cross-document event co-reference are needed. Finally, the timeline extraction systems discussed only utilize information available in the unstructured texts being processed, but the EHR has a wealth of structured information as well. Some timeline systems (not included in this review due to the absence of NLP methods) utilize this structured information to identify events and times for timeline construction (i.e. they ignore the textual data). Thus, future work is needed in integrating the structured timelines and timelines obtained from clinical texts to augment each other for a more complete and accurate representation of a patient's medical journey.

7.2. End-to-end performance evaluation

It is critically important that timeline extraction methods work well with high accuracy before it is implemented in the clinic and could

influence patient care. Because end-to-end systems are built upon many other sub tasks, all sub tasks must perform equally well before the end-to-end system will perform well. For example, Najafabadipour, et.al. [53] noted that many of the errors incurred in timeline construction were the result of incorrect TLINKs generated in the prior step. Thus, evaluation of the individual parts of a timeline extraction system is vital; however, an overall evaluation of the completed timeline is also needed. Unfortunately, there are a limited number of strategies for evaluating end-to-end timeline extraction systems, and within those that do there is a lack of consensus on evaluation methods.

Of the four end-to-end timeline extraction systems discussed in Section 7 only two have any type of formal evaluation for the resulting timeline [53,54]. In order to evaluate the final timeline, both of these had to obtain manual, clinician-annotated timelines, which is highly time-consuming. Dehghan obtained three clinician-annotated patient timelines that started at the patient's diagnosis and continued for the first 42 months. These were used as the gold-standard for comparison with the automatically generated timelines on two levels, first at a high level abstracted to the clinical process such as "Oncology Treatment", and then at a more detailed level that included information like the type of chemotherapy. Additionally, Dehghan binned events (both at the high and low levels of annotation) into 6-month periods where relative order within a window was ignored. Precision and Recall were calculated based on this information and errors were thrown when there were any information mismatches for a given time bin. Najafabadipour et.al. was able to obtain over 800 clinician-annotated timelines for use as a gold-standard, which also started at the date of patient diagnosis. Events are at a low level as compared to Dehghan, and errors were generated if 1) the system identified a different total number of events for a patient, or 2) the same event instance had different date expressions from those identified by the experts. Note that in both of these evaluation strategies, temporal order is either abstracted to bins or ignored all together. While not a complete end-to-end clinical timeline extraction system as defined in this work, Raghavan [116] took a different direction and implemented an edit distance metric based on the popular "word error rate" metric used in automated speech recognition to evaluate how far away an automatically generated timeline is from the gold standard. This method focuses on the sequence of events, and includes the adding, substituting, and deleting of ordered medical events; however, it does not utilize event properties or values such as tumor stage, and only looks at relative order while ignoring durations. An evaluation of a patient timeline should consider both content (to a level of detail useful to clinicians), event order, and event duration. Note that event order does not necessarily mean exact dates, but rather relative ordering. We exclude getting the dates exactly correct as this is a challenging task even for humans; however, the system should be able to infer relative order of unique events and assign a reasonable date of occurrence if one is not explicitly defined in the EHR.

8. Conclusion

In this work, we reviewed the current state of temporal reasoning with respect to timeline extraction in the clinical domain. While much progress has been made, the current state-of-the-art still has a ways to go before practical application in the clinical setting will be possible. We believe there several areas of research that are necessary to make this possible. First, the correct and complete identification of temporal expressions is fundamental to determining when events happened and for placement on a timeline. Temporal expression taggers must be able to identify all types of temporal expressions, including relative, vague, and implicit expressions. In addition, systems need to be able to normalize these expressions to a point on the timeline. This is difficult with relative, vague and implicit TIMEXs, and will require some sort of integrated approach that includes event identification, co-reference resolution, and temporal relation classification in order to determine where an event occurred in a patient's history. A second area of needed work is in developing temporal relation identification systems that perform well on inter-sentence relations and can control for event ordering conflicts. This may require a paradigm shift from looking at pair-wise relations to another framework, such as ranked lists, to control for this issue and reduce computational complexity. Third, constructing a patient's timeline over their entire medical history will require the processing of multiple types of documents, which will have duplicated information. Clinical temporal reasoning systems will need to be able to process this redundant data, which will include cross-document co-reference resolution, in order to limit displaying duplicated events to a clinician. Current work in this area is limited and has room for much progress. Fourth, visualization tools will need to be interoperatable and be able to integrate multiple data types for ease of use by clinicians. Another area of improvement is the development of consistent evaluation methods for timeline extraction systems as a whole so that they can be more easily compared and evaluated. Finally, timeline extraction systems should be able to integrate both structured and unstructured data into the timeline creation process. There are many tools that just use structured data, however, there is much information hidden in the unstructured texts. Being able to integrate this data will be of great benefit to future timeline extraction systems for clinical data.

In conclusion, temporal reasoning over clinical texts has come a long way since the first clinical temporal challenges, however, there is still room for improvement before these systems will be useful to clinicians. Because a patient's medical history is buried in multiple notes with multiple note types and grammar that is not always going to follow traditional rules, future timeline extraction systems should be flexible in processing this diverse data, as well as able to deal with the high level of redundancy in the EHR by integrating this data into a single contiguous timeline through robust co-reference resolution. As the field moves towards annotating the more difficult temporal information, such as relative and implicit temporal expressions, new methods that integrate the normalization of temporal expressions with temporal relation identification and co-reference resolution will be needed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] C.M. MD, Does your child need a tonsillectomy?, 2018. https://www.health.harvard.edu/blog/child-need-tonsillectomy-2018032013504.
[2] A. Ledesma, N. Bidargaddi, J. Strobel, G. Schrader, H. Nieminen, I. Korhonen, M. Ermes, Health timeline: an insight-based study of a timeline visualization of clinical data, BMC Med. Inform. Decis. Mak. 19 (2019) 170.
[3] J.L. Belden, P. Wegier, J. Patel, A. Hutson, C. Plaisant, J.L. Moore, N.J. Lowrance, S.A. Boren, R.J. Koopman, Designing a medication timeline for patients and physicians, J. Am. Med. Inform. Assoc. 26 (2019) 95–105.
[4] A.A.T. Bui, D.R. Aberle, H. Kangarloo, TimeLine: Visualizing Integrated Patient Records, IEEE Trans. Inf Technol. Biomed. 11 (2007) 462–473.
[5] J.S. Hirsch, J.S. Tanenbaum, S. Lipsky Gorman, C. Liu, E. Schmitz, D. Hashorva, A. Ervits, D. Vawdrey, M. Sturm, N. Elhadad, HARVEST, a longitudinal patient record summarizer, J. Am. Med. Informatics Assoc. 22 (2015) 263–274. Publisher: Oxford Academic.
[6] B.S. Glicksberg, B. Oskotsky, P.M. Thangaraj, N. Giangreco, M.A. Badgeley, K.W. Johnson, D. Datta, V.A. Rudrapatna, N. Rappoport, M.M. Shervey, R. Miotto, T.C. Goldstein, E. Rutenberg, R. Frazier, N. Lee, S. Israni, R. Larsen, B. Percha, L. Li, J. T. Dudley, N.P. Tatonetti, A.J. Butte, PatientExploreR: an extensible application for dynamic visualization of patient clinical history from electronic health records in the OMOP common data model, Bioinformatics 35 (2019) 4515–4518. Publisher: Oxford Academic.
[7] D. Reichert, D. Kaufman, B. Bloxham, H. Chase, N. Elhadad, Cognitive Analysis of the Summarization of Longitudinal Patient Records, AMIA Annu. Symp. Proc. 2010 (2010) 667–671.
[8] L. Zhou, G. Hripcsak, Temporal reasoning with medical data-A review with emphasis on medical natural language processing, J. Biomed. Inform. 40 (2007) 183–202.
[9] A. Leeuwenberg, M.-F. Moens, A survey on temporal reasoning for temporal information extraction from text, J. Artif. Intell. Res. 66 (2019) 341–380.

[10] N. Gupta, Temporal information extraction extracting events and temporal expressions a literature survey (2015) 34.
[11] A. Pani, G. Bhattacharjee, Temporal representation and reasoning in artificial intelligence: A review, Math. Comput. Modell. 34 (2001) 55–80.
[12] C.-G. Lim, Y.-S. Jeong, H.-J. Choi, Survey of temporal information extraction, J. Inf. Process. Syst. 15 (2019) 931–956.
[13] W. Sun, A. Rumshisky, O. Uzuner, Evaluating temporal relations in clinical text: 2012 i2b2 Challenge, J. Am. Med. Informatics Assoc.: JAMIA 20 (2013) 806–813.
[14] S. Bethard, L. Derczynski, G. Savova, J. Pustejovsky, M. Verhagen, SemEval-2015 Task 6: Clinical TempEval, in: Proceedings of the 9th International Workshop on Semantic Evaluation (SemEval 2015), Association for Computational Linguistics, Denver, Colorado, 2015, pp. 806–814.
[15] W.F.S. Iv, S. Bethard, S. Finan, M. Palmer, S. Pradhan, P.C. d. Groen, B. Erickson, T. Miller, C. Lin, G. Savova, J. Pustejovsky, Temporal annotation in the clinical domain, Trans. Assoc. Comput. Linguist. 2 (2014) 143–154. Number: 0.
[16] J. Strötgen, M. Gertz, Heideltime: High quality rule-based extraction and normalization of temporal expressions, in: Proceedings of the 5th International Workshop on Semantic Evaluation, SemEval '10, Association for Computational Linguistics, Stroudsburg, PA, USA, 2010, pp. 321–324.
[17] A.X. Chang, C.D. Manning, Sutime: A library for recognizing and normalizing time expressions., in: Lrec, vol. 2012, 2012, pp. 3735–3740.
[18] W. Sun, A. Rumshisky, O. Uzuner, Temporal reasoning over clinical text: the state of the art, J. Am. Med. Inform. Assoc. 20 (2013) 814–819.
[19] J.C. Augusto, Temporal reasoning for decision support in medicine, Artif. Intell. Med. 33 (2005) 1–24.
[20] M. Stacey, C. McGregor, Temporal abstraction in intelligent clinical data analysis: A survey, Artif. Intell. Med. 39 (2007) 1–24.
[21] L. Ferro, I. Mani, B. Sundheim, G. Wilson, TIDES Temporal Annotation Guidelines Version 1.0.2 (2001) 57.
[22] J. Pustejovsky, J.M. Castano, R. Ingria, R. Sauri, R.J. Gaizauskas, A. Setzer, G. Katz, D.R. Radev, Timeml: Robust specification of event and temporal expressions in text, New Direct. Question Answering 3 (2003) 28–34.
[23] A. Setzer, Temporal information in newswire articles: an annotation scheme and corpus study., phd, University of Sheffield, 2002 http://etheses.whiterose.ac.uk/14436/.
[24] E. Saquete, J. Pustejovsky, Automatic transformation from TIDES to TimeML annotation, Lang. Resour. Eval. 45 (2011) 495–523.
[25] J. Pustejovsky, K. Lee, H. Bunt, L. Romary, ISO-TimeML: An International Standard for Semantic Annotation, in: N. Calzolari (Ed.), Proceedings of the Seventh International Confer- ence on Language Resources and Evaluation (LREC'10), Valletta, Malta, 2010, p. 4.
[26] S. Bethard, J. Parker, A semantically compositional annotation scheme for time normalization, in: Lrec, volume 2016, 2016, pp. 3779–3786.
[27] R. Grishman, B. Sundheim, Design of the MUC-6 Evaluation, in: Sixth Message Understanding Conference (MUC-6), in: Sixth Message Understanding Conference (MUC-6): Proceedings of a Conference Held in Columbia, Maryland, November 6-8, 1995, 1995 https://www.aclweb.org/anthology/M95-1001.
[28] B. Onyshkevych, M.E. Okurowski, L. Carlson, Tasks, Domains, and Languages, in: Fifth Message Understanding Conference (MUC-5): Proceedings of a Conference Held in Baltimore, Maryland, August 25–27, 1993, 1993. https://www.aclweb.org/anthology/M93-1002.
[29] N.A. Chinchor, Overview of MUC-7, in: Seventh Message Understanding Conference (MUC-7): Proceedings of a Conference Held in Fairfax, Virginia, April 29 – May 1, 1998, 1998. https://www.aclweb.org/anthology/M98-1001.
[30] M. Negri, L. Marseglia, Recognition and Normalization of Time Expressions: ITC-irst at TERN 2004, 2004.
[31] M. Verhagen, R. Gaizauskas, F. Schilder, M. Hepple, G. Katz, J. Pustejovsky, SemEval-2007 Task 15: TempEval Temporal Relation Identification, in: Proceedings of the 4th International Workshop on Semantic Evaluations, SemEval '07, Association for Computational Linguistics, Stroudsburg, PA, USA, 2007, pp. 75–80. http://dl.acm.org/citation.cfm?id=1621474.1621488.
[32] M. Verhagen, R. Saurí, T. Caselli, J. Pustejovsky, SemEval-2010 Task 13: TempEval-2, in: Proceedings of the 5th International Workshop on Semantic Evaluation, SemEval '10, Association for Computational Linguistics, Stroudsburg, PA, USA, 2010, pp. 57–62. http://dl.acm.org/citation.cfm?id=1859664.1859674.
[33] N. UzZaman, H. Llorens, L. Derczynski, J. Allen, M. Verhagen, J. Pustejovsky, SemEval-2013 Task 1: TempEval-3: Evaluating Time Expressions, Events, and Temporal Relations, in: Second Joint Conference on Lexical and Computational Semantics (*SEM), Volume 2: Proceedings of the Seventh International Workshop on Semantic Evaluation (SemEval 2013), Association for Computational Linguistics, Atlanta, Georgia, USA, 2013, pp. 1–9.
[34] A.E.W. Johnson, T.J. Pollard, L. Shen, L.-W. H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. Anthony Celi, R.G. Mark, MIMIC-III, a freely accessible critical care database, Sci. Data 3 (2016) 160035. Number: 1 Publisher: Nature Publishing Group.
[35] Y. Cheng, P. Anick, P. Hong, N. Xue, Temporal relation discovery between events and temporal expressions identified in clinical narrative, J. Biomed. Inform. 46 (2013) S48–S53.
[36] J. D'Souza, V. Ng, Knowledge-rich temporal relation identification and classification in clinical notes, Database 2014 (2014a). Publisher: Oxford Academic.
[37] J. D'Souza, V. Ng, Annotating Inter-Sentence Temporal Relations in Clinical Notes (2014b) 8.
[38] W. Wang, K. Kreimeyer, E.J. Woo, R. Ball, M. Foster, A. Pandey, J. Scott, T. Botsis, A new algorithmic approach for the extraction of temporal associations from clinical narratives with an application to medical product safety surveillance reports, J. Biomed. Inform. 62 (2016) 78–89.
[39] H.-J. Lee, Y. Zhang, J. Xu, C. Tao, H. Xu, M. Jiang, Towards practical temporal relation extraction from clinical notes: An analysis of direct temporal relations, in: 2017 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2017, pp. 1272–1275, https://doi.org/10.1109/BIBM.2017.8217842.
[40] H.-J. Lee, Y. Zhang, M. Jiang, J. Xu, C. Tao, H. Xu, Identifying direct temporal relations between time and events from clinical notes, BMC Med. Inform. Decis. Mak. 18 (2018) 49.
[41] H. Guan, J. Li, H. Xu, M. Devarakonda, Robustly Pre-trained Neural Model for Direct Temporal Relation Extraction, arXiv:2004.06216 [cs] (2020). ArXiv: 2004.06216.
[42] C. Lin, D. Dligach, T.A. Miller, S. Bethard, G.K. Savova, Multilayered temporal modeling for the clinical domain, J. Am. Med. Inform. Assoc. 23 (2016) 387–395. Publisher: Oxford Academic.
[43] R. Patel, S. Tanwani, Temporal relation identification from clinical text using LSTM based deep learning model 5 (2018) 5.
[44] S. Bethard, G. Savova, W.-T. Chen, L. Derczynski, J. Pustejovsky, M. Verhagen, SemEval-2016 Task 12: Clinical TempEval, in: Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016), Association for Computational Linguistics, San Diego, California, 2016, pp. 1052–1062. http://www.aclweb.org/anthology/S16-1165.
[45] S. Bethard, G. Savova, M. Palmer, J. Pustejovsky, SemEval-2017 Task 12: Clinical TempEval, in: Proceedings of the 11th International Workshop on Semantic Evaluation (SemEval-2017), Association for Computational Linguistics, Vancouver, Canada, 2017, pp. 565–572. http://aclweb.org/anthology/S17-2093. doi:10.18653/v1/S17-2093.
[46] E. Laparra, D. Xu, S. Bethard, A.S. Elsayed, M. Palmer, Semeval 2018 task 6: Parsing time normalization, in: Proceedings of the 12th International Workshop on Semantic Evaluation, SemEval '18, Association for Computational Linguistics, New Orleans, LA, USA, 2018.
[47] A. Olex, L. Maffey, N. Morgan, B. McInnes, Chrono at SemEval-2018 Task 6: A System for Normalizing Temporal Expressions, in: Proceedings of The 12th International Workshop on Semantic Evaluation, Association for Computational Linguistics, New Orleans, Louisiana, 2018, pp. 97–101, http://www.aclweb.org/anthology/S18-1012.
[48] S. Bethard, ClearTK-TimeML: A minimalist approach to TempEval 2013, in: Second Joint Conference on Lexical and Computational Semantics (*SEM), Volume 2: Proceedings of the Seventh International Workshop on Semantic Evaluation (SemEval 2013), Association for Computational Linguistics, Atlanta, Georgia, USA, 2013, pp. 10–14. http://www.aclweb.org/anthology/S13-2002.
[49] S. Sohn, K.B. Wagholikar, D. Li, S.R. Jonnalagadda, C. Tao, R. Komandur Elayavilli, H. Liu, Comprehensive temporal information detection from clinical text: medical events, time, and TLINK identification, J. Am. Med. Inform. Assoc. 20 (2013) 836–842.
[50] A. Olex, L. Maffey, B. McInnes, NLP Whack-A-Mole: Challenges in Cross-Domain Temporal Expression Extraction, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), Association for Computational Linguistics, Minneapolis, Minnesota, 2019, pp. 3682–3692. https://www.aclweb.org/anthology/N19-1369. doi:10.18653/v1/N19-1369.
[51] R. Sarkar, B. Nayal, A. Joshi, An Approach for Temporal Ordering of Medical Case Reports, in: V.E. Balas, N. Sharma, A. Chakrabarti (Eds.), Data Management, Analytics and Innovation, Advances in Intelligent Systems and Computing, Springer, Singapore, 2019, pp. 131–141.
[52] M. Najafabadipour, J.M. Tuñas, A. Rodríguez-González, E. Menasalvas, Analysis of Electronic Health Records to Identify the Patient's Treatment Lines: Challenges and Opportunities, in: M. Bramer, M. Petridis (Eds.), Artificial Intelligence XXXVI, Lect. Notes Comput. Sci., Springer International Publishing, Cham, 2019, pp. 437–442, https://doi.org/10.1007/978-3-030-34885-4_33.
[53] M. Najafabadipour, M. Zanin, A. Rodríguez-González, M. Torrente, B. Nuñez García, J.L. Cruz Bermudez, M. Provencio, E. Menasalvas, Reconstructing the patient's natural history from electronic health records, Artif. Intell. Med. 105 (2020) 101860.
[54] A. Dehghan, Mining Patient Journeys from Healthcare Narratives, 2014. https://www.research.manchester.ac.uk/portal/files/54570636/FULL_TEXT.PDF.
[55] A. Dehghan, Temporal ordering of clinical events, arXiv:1504.03659 [cs] (2015). ArXiv: 1504.03659.
[56] J. Tourille, Extracting Clinical Event Timelines: Temporal Information Extraction and Coreference Resolution in Electronic Health Records, phdthesis, Université Paris-Saclay, 2018. https://tel.archives-ouvertes.fr/tel-01997223.
[57] H.-J. Lee, H. Xu, J. Wang, Y. Zhang, S. Moon, J. Xu, Y. Wu, UTHealth at SemEval-2016 Task 12: an End-to-End System for Temporal Information Extraction from Clinical Notes, in: Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016), Association for Computational Linguistics, San Diego, California, 2016, pp. 1292–1297, https://doi.org/10.18653/v1/S16-1201, https://www.aclweb.org/anthology/S16-1201.
[58] O. Uzuner, B.R. South, S. Shen, S.L. DuVall, i2b2/VA challenge on concepts, assertions, and relations in clinical text, J. Am. Med. Inform. Assoc. 18 (2011) (2010) 552–556.
[59] N. Viani, J. Kam, L. Yin, S. Verma, R. Stewart, R. Patel, S. Velupillai, Annotating Temporal Relations to Determine the Onset of Psychosis Symptoms, Stud. Health Technol. Inform. 264 (2019) 418–422.
[60] J.O. Wrenn, D.M. Stein, S. Bakken, P.D. Stetson, Quantifying clinical narrative redundancy in an electronic health record, J. Am. Med. Inform. Assoc. 17 (2010) 49–53.

[61] Y. Xu, Y. Wang, T. Liu, J. Tsujii, E.I.-C. Chang, An end-to-end system to identify temporal relation in discharge summaries: 2012 i2b2 challenge, J. Am. Med. Inform. Assoc. 20 (2013) 849–858.
[62] C. Grouin, N. Grabar, T. Hamon, S. Rosset, X. Tannier, P. Zweigenbaum, Eventual situations for timeline extraction from clinical reports, J. Am. Med. Inform. Assoc. 20 (2013) 820–827.
[63] J. D'Souza, V. Ng, Temporal Relation Identification and Classification in Clinical Notes, in: Proceedings of the International Conference on Bioinformatics, Computational Biology and Biomedical Informatics, Association for Computing Machinery, Wshington DC, USA, 2013, pp. 392–401, https://doi.org/10.1145/2506583.2506654.
[64] Y.-C. Chang, H.-J. Dai, J.C.-Y. Wu, J.-M. Chen, R.T.-H. Tsai, W.-L. Hsu, TEMPTING system: A hybrid method of rule and machine learning for temporal relation extraction in patient discharge summaries, J. Biomed. Inform. 46 (2013) S54–S62.
[65] B. Tang, Y. Wu, M. Jiang, Y. Chen, J.C. Denny, H. Xu, A hybrid system for temporal information extraction from clinical text, J. Am. Med. Inform. Assoc. 20 (2013) 828–835.
[66] D. Dligach, T. Miller, C. Lin, S. Bethard, G. Savova, Neural Temporal Relation Extraction, in: Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers, Association for Computational Linguistics, Valencia, Spain, 2017, pp. 746–751. http://www.aclweb.org/anthology/E17-2118.
[67] J. Tourille, O. Ferret, A. Névéol, X. Tannier, Neural Architecture for Temporal Relation Extraction: A Bi-LSTM Approach for Detecting Narrative Containers, in: Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), Association for Computational Linguistics, Vancouver, Canada, 2017, pp. 224–230. https://www.aclweb.org/anthology/P17-2035. doi:10.18653/v1/P17-2035.
[68] C. Lin, T. Miller, D. Dligach, H. Amiri, S. Bethard, G. Savova, Self-training improves Recurrent Neural Networks performance for Temporal Relation Extraction, in: Proceedings of the Ninth International Workshop on Health Text Mining and Information Analysis, Association for Computational Linguistics, Brussels, Belgium, 2018, pp. 165–176. https://www.aclweb.org/anthology/W18-5619. doi:10.18653/v1/W18-5619.
[69] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), Association for Computational Linguistics, Minneapolis, Minnesota, 2019, pp. 4171–4186. https://www.aclweb.org/anthology/N19-1423. doi:10.18653/v1/N19-1423.
[70] Z. Liu, Y. Lin, M. Sun, Representation Learning and NLP, in: Z. Liu, Y. Lin, M. Sun (Eds.), Representation Learning for Natural Language Processing, Springer, Singapore, 2020, pp. 1–11, https://doi.org/10.1007/978-981-15-5573-2_1.
[71] T. Mikolov, I. Sutskever, K. Chen, G.S. Corrado, J. Dean, Distributed Representations of Words and Phrases and their Compositionality, in: C.J.C. Burges, L. Bottou, M. Welling, Z. Ghahramani, K.Q. Weinberger (Eds.), Advances in Neural Information Processing Systems 26, Curran Associates Inc, 2013, pp. 3111–3119. http://papers.nips.cc/paper/5021-distributed-representations-of-words-and-phrases-and-their-compositionality.pdf.
[72] J. Pennington, R. Socher, C. Manning, Glove: Global Vectors for Word Representation, in: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics, Doha, Qatar, 2014, pp. 1532–1543. https://www.aclweb.org/anthology/D14-1162. doi:10.3115/v1/D14-1162.
[73] M. Peters, M. Neumann, M. Iyyer, M. Gardner, C. Clark, K. Lee, L. Zettlemoyer, Deep Contextualized Word Representations, in: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers), Association for Computational Linguistics, New Orleans, Louisiana, 2018, pp. 2227–2237. https://www.aclweb.org/anthology/N18-1202. doi:10.18653/v1/N18-1202.
[74] A. Radford, K. Narasimhan, T. Salimans, I. Sutskever, Improving Language Understanding by Generative Pre-Training, Technical Report (2018) 12.
[75] C. Lin, T. Miller, D. Dligach, S. Bethard, G. Savova, A BERT-based Universal Model for Both Within- and Cross-sentence Clinical Temporal Relation Extraction, in: Proceedings of the 2nd Clinical Natural Language Processing Workshop, Association for Computational Linguistics, Minneapolis, Minnesota, USA, 2019, pp. 65–71. https://www.aclweb.org/anthology/W19-1908. doi:10.18653/v1/W19-1908.
[76] C. Lin, T. Miller, D. Dligach, F. Sadeque, S. Bethard, G. Savova, A BERT-based One-Pass Multi-Task Model for Clinical Temporal Relation Extraction, in: Proceedings of the 19th SIGBioMed Workshop on Biomedical Language Processing, Association for Computational Linguistics, 2020, pp. 70–75.
[77] H. Wang, M. Tan, M. Yu, S. Chang, D. Wang, K. Xu, X. Guo, S. Potdar, Extracting Multiple-Relations in One-Pass with Pre-Trained Transformers, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, Florence, Italy, 2019, pp. 1371–1377, https://doi.org/10.18653/v1/P19-1132, https://www.aclweb.org/anthology/P19-1132.
[78] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, V. Stoyanov, RoBERTa: A Robustly Optimized BERT Pretraining Approach, arXiv:1907.11692 [cs] (2019).
[79] R. Sukthanker, S. Poria, E. Cambria, R. Thirunavukarasu, Anaphora and coreference resolution: A review, Inf. Fusion 59 (2020) 139–162.
[80] R. Zhang, S. Pakhomov, B.T. McInnes, G.B. Melton, Evaluating measures of redundancy in clinical texts, AMIA Annu. Symp. Proc. 2011 (2011) 1612–1620.
[81] O. Uzuner, A. Bodnari, S. Shen, T. Forbush, J. Pestian, B.R. South, Evaluating the state of the art in coreference resolution for electronic medical records, J. Am. Med. Inform. Assoc. 19 (2012) 786–791.
[82] O. Bodenreider, The Unified Medical Language System (UMLS): integrating biomedical terminology, Nucleic Acids Res. 32 (2004) D267–D270.
[83] Y. Xu, J. Liu, J. Wu, Y. Wang, Z. Tu, J.-T. Sun, J. Tsujii, E.I.-C. Chang, A classification approach to coreference in discharge summaries: 2011 i2b2 challenge, J. Am. Med. Inform. Assoc. 19 (2012) (2011) 897–905.
[84] D. Hinote, C. Ramirez, P. Chen, A comparative study of co-refernece resolution in clinical text, in: Proceedings of the 2011 i2b2/VA/Cincinnati Workshop on Challenges in Natural Language Processing for Clinical Data, Boston, MA, USA, 2011.
[85] P. Anick, P. Hong, N. Xue, Y. Yang, Coreference resolution for electronic medical records, in: Proceedings of the 2011 i2b2/VA/Cincinnati Workshop on Challenges in Natural Language Processing for Clinical Data, Boston, MA, USA, 2011.
[86] P. Raghavan, E. Fosler-Lussier, N. Elhadad, A.M. Lai, Cross-narrative Temporal Ordering of Medical Events, in: Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Baltimore, Maryland, 2014, pp. 998–1008. https://www.aclweb.org/anthology/P14-1094. doi:10.3115/v1/P14-1094.
[87] R. Sukthanker, S. Poria, E. Cambria, R. Thirunavukarasu, Anaphora and Coreference Resolution: A Review, arXiv:1805.11824 [cs] (2018). ArXiv: 1805.11824.
[88] A. Dehghan, A. Kovacevic, G. Karystianis, J.A. Keane, G. Nenadic, Combining knowledge- and data-driven methods for de-identification of clinical narratives, J. Biomed. Inform. 58 (2015) S53–S59.
[89] P. Bramsen, P. Deshpande, Y.K. Lee, R. Barzilay, Inducing Temporal Graphs, in: Proceedings of the 2006 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Sydney, Australia, 2006, pp. 189–198, https://www.aclweb.org/anthology/W06-1623.
[90] P. Raghavan, E. Fosler-Lussier, A.M. Lai, Exploring Semi-supervised Coreference Resolution of Medical Concepts Using Semantic and Temporal Features, in: Proceedings of the 2012 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL HLT '12, Association for Computational Linguistics, Stroudsburg, PA, USA, 2012, pp. 731–741. http://dl.acm.org/citation.cfm?id=2382029.2382148.
[91] S. Jeblee, G. Hirst, Listwise temporal ordering of events in clinical notes, in: Proceedings of the Ninth International Workshop on Health Text Mining and Information Analysis, Association for Computational Linguistics, Brussels, Belgium, 2018, pp. 177–182. https://www.aclweb.org/anthology/W18-5620. doi:10.18653/v1/W18-5620.
[92] S. Martins, Y. Shahar, M. Galperin, H. Kaizer, D. Goren-bar, Evaluation of KNAVE-II: a Tool for Intelligent Query and Exploration of Patient Data, 2004.
[93] D. Klimov, Y. Shahar, M. Taieb-Maimon, Intelligent visualization and exploration of time-oriented data of multiple patients, Artif. Intell. Med. 49 (2010) 11–31.
[94] D. Gotz, F. Wang, A. Perer, A methodology for interactive mining and visual analysis of clinical event patterns using electronic health record data, J. Biomed. Inform. 48 (2014) 148–159.
[95] H. Park, J. Choi, V-model: a new innovative model to chronologically visualize narrative clinical texts, in: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, CHI '12, Association for Computing Machinery, New York, NY, USA, 2012, pp. 453–462. doi: 10.1145/2207676.2207739.
[96] A. Nikfarjam, E. Emadzadeh, G. Gonzalez, Towards generating a patient's timeline: Extracting temporal relationships from clinical notes, J. Biomed. Inform. Suppl. (2013) S40–S47.
[97] A. Maharana, Extraction of Clinical Timeline from Discharge Summaries using Neural Networks, Thesis, 2017. https://digital.lib.washington.edu:443/researchworks/handle/1773/40821, accepted: 2018-01-20T00:57:43Z.
[98] C. Hallett, Multi-modal presentation of medical histories, in: Proceedings of the 13th international conference on Intelligent user interfaces, IUI '08, Association for Computing Machinery, Gran Canaria, Spain, 2008, pp. 80–89. doi: 10.1145/1378773.1378785.
[99] J. Rogers, C. Puleston, A. Rector, The CLEF Chronicle: Patient Histories Derived from Electronic Health Records, in: 22nd International Conference on Data Engineering Workshops (ICDEW'06), 2006, p. x109, https://doi.org/10.1109/ICDEW.2006.144.
[100] H. Harkema, I. Roberts, R. Gaizauskas, M. Hepple, Information-Extraction-from-Clinical-Records.pdf, 2005. https://www.researchgate.net/profile/M_Hepple/publication/246471156_Information_Extraction_from_Clinical_Records/links/53e9d2720cf2dc24b3cad9bf/Information-Extraction-from-Clinical-Records.pdf.
[101] Z. Yuan, S. Finan, J. Warner, G. Savova, H. Hochheiser, Interactive Exploration of Longitudinal Cancer Patient Histories Extracted From Clinical Text, JCO Clin. Cancer Inform. (2020) 412–420. Publisher: American Society of Clinical Oncology.
[102] E. Laparra, I. Aldabe, G. Rigau, Document Level Time-anchoring for TimeLine Extraction, in: Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 2: Short Papers), Association for Computational Linguistics, Beijing, China, 2015, pp. 358–364. http://aclweb.org/anthology/P15-2059. doi:10.3115/v1/P15-2059.
[103] S. Cornegruta, A. Vlachos, Timeline extraction using distant supervision and joint inference, in: Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Austin,

Texas, 2016, pp. 1936–1942, https://doi.org/10.18653/v1/D16-1200, https://www.aclweb.org/anthology/D16-1200.
[104] E. Saquete Boró, B. Navarro Colorado, Cross-Document Event Ordering through Temporal Relation Inference and Distributional Semantic Models (2017).
[105] E. Laparra, R. Agerri, I. Aldabe, G. Rigau, Multi-lingual and Cross-lingual timeline extraction, Knowl.-Based Syst. 133 (2017) 77–89.
[106] L. Zhou, C. Friedman, S. Parsons, G. Hripcsak, System Architecture for Temporal Information Extraction, Representation and Reasoning in Clinical Narrative Reports, AMIA Annu. Symp. Proc. 2005 (2005) 869–873.
[107] H. Jung, J. Allen, N. Blaylock, W. de Beaumont, L. Galescu, M. Swift, Building Timelines from Narrative Clinical Records: Initial Results Based-on Deep Natural Language Understanding, in: Proceedings of BioNLP 2011 Workshop, BioNLP '11, Association for Computational Linguistics, Stroudsburg, PA, USA, 2011, pp. 146–154. http://dl.acm.org/citation.cfm?id=2002902.2002924.
[108] J. Allen, D. Byron, M. Dzikovska, G. Ferguson, L. Galescu, A. Stent, An architecture for a generic dialogue shell, Natural Lang. Eng. 6 (2000) 213–228. Publisher: Cambridge University Press.
[109] M. Verhagen, Temporal Closure in an Annotation Environment, Lang. Resour. Eval. 39 (2005) 211–241. Publisher: Springer.
[110] W.W. Cohen, P. Ravikumar, S.E. Fienberg, A Comparison of String Distance Metrics for Name-Matching Tasks, 2003, pp. 73–78.
[111] D. Ferrucci, A. Lally, UIMA: an architectural approach to unstructured information processing in the corporate research environment, Natural Lang. Eng. 10 (2004) 327–348.
[112] E. Menasalvas Ruiz, J.M. Tuñas, G. Bermejo, C. Gonzalo Martín, A. Rodríguez-González, M. Zanin, C. González de Pedro, M. Méndez, O. Zaretskaia, J. Rey, C. Parejo, J.L. Cruz Bermudez, M. Provencio, Profiling Lung Cancer Patients Using Electronic Health Records, J. Med. Syst. 42 (2018) 126.
[113] M. Najafabadipour, J.M. Tuñas, A. Rodríguez-González, E. Menasalvas, Lung Cancer Concept Annotation from Spanish Clinical Narratives, in: S. Auer, M.-E. Vidal (Eds.), Data Integration in the Life Sciences, Lect. Notes Comput. Sci., Springer International Publishing, Cham, 2019, pp. 153–163, https://doi.org/10.1007/978-3-030-06016-9_15.
[114] M. Najafabadipour, M. Zanin, A. Rodríguez-González, C. Gonzalo-Martín, B.N. García, V. Calvo, J.L.C. Bermudez, M. Provencio, E. Menasalvas, Recognition of Time Expressions in Spanish Electronic Health Records, in: 2019 IEEE 32nd International Symposium on Computer-Based Medical Systems (CBMS), 2019b, pp. 69–74. doi:10.1109/CBMS.2019.00025, iSSN: 2372-9198.
[115] M. Straka, J. Hajič, J. Straková, UDPipe: Trainable Pipeline for Processing CoNLL-U Files Performing Tokenization, Morphological Analysis, POS Tagging and Parsing, in: Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16), European Language Resources Association (ELRA), Portoro++, Slovenia, 2016, pp. 4290–4297. https://www.aclweb.org/anthology/L16-1680.
[116] P. Raghavan, Medical Event Timeline Generation from Clinical Narratives, Ph.D. thesis, The Ohio State University, 2014. https://etd.ohiolink.edu/pg_10?0::NO:10:P10_ACCESSION_NUM:osu1397651496.